ANSI/IES LM-10-20(R2023)

Illuminating ENGINEERING SOCIETY

APPROVED METHOD: PHOTOMETRIC TESTING OF ROADWAY AND AREA LIGHTING FLUORESCENT LUMINAIRES

AN AMERICAN NATIONAL STANDARD

Approved American National Standard ANSI

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Publication of this document has been approved by the IES. Suggestions for revision should be directed to the IES.

Prepared by
The IES Testing Procedures Committee

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Foreword

This approved testing methods guide is an update of LM-10-1996. The guide has been updated to coordinate with *ANSI/IES LM-31-20, Approved Method: Photometric Testing of Roadway Luminaires Using High Intensity Discharge and Incandescent Filament Lamps*, and *ANSI/IES LM-41-20, Approved Method: Photometric Testing of Indoor Fluorescent Luminaires*.[1]

1.0 Introduction and Scope

1.1 Introduction

Exterior illumination provided by luminaires generally falls into the standard Type I, II, III, IV, V and VS classifications. Distribution patterns are typically bisymmetric, quadrilaterally symmetric, or completely symmetric. Illumination provided by luminaires such as outdoor wall packs, bollards, and unique cases of general area lighting may not be designed to fit into the standard classifications and may have completely asymmetric distribution patterns.

The test procedures for luminaires using fluorescent lamps described herein can utilize either absolute or relative photometry methods. In both cases, general conditions of suitability such as test distance should be assessed for adequacy to ensure the validity of the test results.

1.2 Scope

This Lighting Measurement (LM) guide defines adequate and uniform methods for measuring and reporting the photometric characteristics of roadway and area lighting fluorescent luminaires. It describes characteristics of these luminaires and some components, as well as the requirements for the thermal environment and proper control of the electrical and mechanical systems involved. General test conditions and the testing procedure best suited for achieving accurate and consistent photometric results are defined.

2.0 Normative References

2.1 ANSI/IES LM-9-20(R23)

Illuminating Engineering Society. Approved Method: Electrical and Photometric Measurements of Fluorescent Lamps. New York: IES; 2020.

2.2 ANSI/IES LM-28-20

Illuminating Engineering Society. Approved Method: Guide for the Selection, Care and Use of Electrical Instruments in the Photometric Laboratory. New York: IES; 2020.

2.3 ANSI/IES LM-54-20

Illuminating Engineering Society. Approved Method: Guide to Lamp Seasoning. New York: IES; 2020.

2.4 ANSI/IES LM-66-20(R23)

Illuminating Engineering Society. Approved Method: Electrical and Photometric Measurements of Single Based Fluorescent Lamps. New York: The Society; 2020.

2.5 ANSI/IES LS-1-20

Illuminating Engineering Society. Nomenclature and Definitions for Illuminating Engineering. New York: IES; 2020. Online: https://www.ies.org/standards/definitions/. (Accessed 2019 Sep 18).

3.0 Nomenclature and Definitions

The terms used in this document follow the definitions given in **Normative Reference 2.5**. Additional terms are defined below.

3.1 area lighting

In this document, *area lighting* refers to exterior illumination provided by luminaires (such as outdoor wall packs, bollards, and general area lighting luminaires) that may not be designed to fit into the standard Type I, II, III, IV, V, or VS classifications.[2]

3.2 pre-burning

Pre-burning is defined as operating a previously seasoned lamp for a sufficient period of time such that it reaches thermal equilibrium, the mercury locates at the coolest zone of the lamp, and a stable light output is achieved under the test conditions.

3.3 test distance

The *test distance* shall be defined as the distance along the light path (direct and/or reflected) from the center of rotation of the goniophotometer axes as shown in **Figure 6-1** (**Section 6.2**), to the surface of the photodetector.

3.4 units

The units of electrical measurement used in this LM are the volt, ampere, and watt. The units of photometric measurement are the lumen and the candela. Color is specified using chromaticity via chromaticity coordinates. 1976 (u', v') chromaticity coordinates are preferred, but 1931 (x, y) chromaticity coordinates may also be used.

4.0 Physical and Environmental Test Conditions

4.1 General

The photometric values and electrical characteristics of fluorescent lamps and luminaires are sensitive to ambient temperature and air movement. In addition, fluorescent lamps and luminaires can be sensitive to operating orientation and physical shock and therefore should be handled with care during both storage and testing in the laboratory. It is good laboratory practice for the storage and testing of lamps to be undertaken in a relatively clean environment.

4.2 Temperature

Referring to **Figure 4-1**, the ambient temperature of the photometric laboratory within 1 meter of the luminaire, in Zone 2, shall be maintained at 25 °C ± 1 °C (77 °F ± 2 °F). The ambient temperature shall be measured at the same height and not more than 1.0 meter (3.3 feet) from the luminaire. The ambient temperature in the goniophotometer room, as shown in **Zone 1**, shall be maintained at 25 °C ± 5 °C (77 °F ± 9 °F). Temperature sensors shall be shielded from direct radiation and convection currents produced by the luminaire.

4.3 Airflow

The incidence of air movement on the surface of a fluorescent product under test may substantially alter electrical and photometric values. Airflow around the fluorescent product under test should be such that normal convective airflow induced by the device under test is not affected.

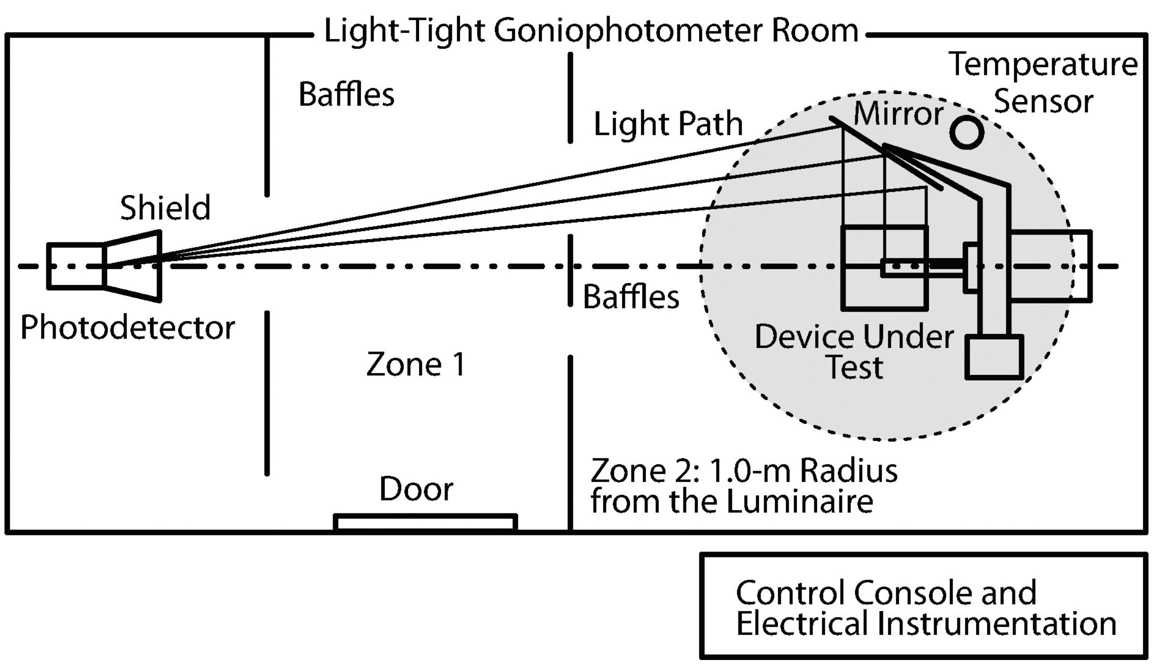


Figure 4-1. Layout of a typical moving-mirror goniophotometer. (© Illuminating Engineering Society)

4.4 Stray Light

Stray light should be suppressed in the test environment through the adequate use of low-reflectance surface finishes, shielding and baffling. Methods for stray light suppression include limiting the field of view, reducing reflected light, and using light traps.[2] The walls, ceiling and floor of the photometric test room should be painted flat black or covered with a diffuse, low-reflectance material such as black felt, black velvet, or black carpeting. It is important to note that the directional response of some materials should also be considered. Opaque low-reflectance baffles with limiting apertures interposed between the photodetector position and the luminaire position that shield the photodetector, except in the direction of the test source, should aid in suppressing the stray light reaching the photodetector. For moving-photodetector goniophotometers, the light incident on the photodetector should only be that which is directly transmitted from the luminaire or bare lamp. For moving-mirror goniophotometers, only the light directly from the luminaire or bare lamp that is reflected via the mirror should be measured. Stray light may be measured by running a complete or partial test with all direct light from the lamp or luminaire shielded from the photodetector. The choice of partial or complete measurement of the stray light depends on the spatial properties of the luminaire's photometric pattern and the expected stray light contribution at each measured vertical and horizontal angle pair. The measured stray light, if any, shall then be subtracted from the bare lamp or luminaire test data, at each vertical angle in each plane measured.

4.5 Vibration

Lamps should not be subjected to excessive vibration or shock during measurement.

5.0 Electrical Test Conditions

5.1 Power Supply Requirements

5.1.1 Primary Source Voltage. Ballasts should be operated at their nominal rated primary voltage and frequency. If the voltage rating is a range, the ballast should be operated at the intended application voltage, and the test voltage shall recorded in the test report.

5.1.2 Power Supply Wave Shape. The AC power supply, while operating the luminaire or test lamp, shall have a sinusoidal voltage wave shape such that the value of the root mean square (RMS) summation of the harmonic components shall not exceed 3% of the fundamental. Caution should be exercised when using constant-voltage transformer regulators. This system type can produce harmonic distortion in excess of 3% when lightly loaded.

5.1.3 Power Supply Regulation. Ballast supply voltage should not vary more than ±0.5% during the test.

5.2 Reference Circuit Requirements

There are no reference circuit requirements.

5.3 Measurement Instrument Requirements

Electrical parameters (e.g., voltage, current, and power) shall be measured with instruments having calibration that is traceable to a national metrological institute (NMI). The instrument measurement tolerance* of alternating current (AC) voltage and current shall be less than or equal to 0.2%. The instrument measurement tolerance of AC power shall be less than or equal to 0.5%. The instrument measurement tolerance of direct current (DC) voltage and current shall be less than or equal to 0.1%.

Instrument measurement ranges should be selected so that the mid-to-maximum portion of the range is used for any specific measurement. Since gaseous discharge lamps may have greatly distorted wave shapes, AC instruments (for voltage, current, and power) shall measure true root-mean-square (RMS) values. Instruments with scales calibrated in RMS values but with deflections or readings based on average or peak values shall not be used. If analog instrumentation is used,

* Expanded uncertainty with a 95% confidence interval, normally with a coverage factor *k*=2, as prescribed in *JCGM* 100:2008, *Evaluation of measurement data - Guide to Expression of Uncertainty in Measurement*, and *ANSI/NCSL Z540-2-1997, U.S. Guide to the Expression of Uncertainty in Measurement*. If a manufacturer's specification does not specify uncertainty this way, then the manufacturer should be contacted for proper conversion.

large-scale deflection for the intended measurement range should be used.

The electrical instruments should be chosen according to the guidelines and precautions presented in IES LM-28-12 (see **Section 2.2**). It should be noted that most electronic ballasts provide a high-frequency lamp current (typically greater than 20,000 Hz). Care should be taken that possible conducted or radiated electromagnetic interference (EMI) from this type of ballast does not cause reading errors. High frequency wire runs should be as short as possible and, if necessary, shielded from EMI. It may be necessary to use precautions in order to prevent EMI from reaching voltage monitoring equipment or the photodetector readout device through the power circuits.

6.0 Testing Procedure Requirements

6.1 General

The test procedures for fluorescent luminaires can utilize either absolute or relative photometry methods. In both cases general conditions of suitability such as test distance should be assessed for adequacy to ensure the validity of the test results. For relative photometry, the key process steps are bare-lamp calibration and luminaire testing. For absolute photometry, the key process steps are absolute system calibration and luminaire testing.

The luminaire shall be operated under conditions as nearly typical of end use as possible, or suitable corrections made. For example, if a luminaire is intended to be mounted on a pole at a tilted angle, it shall be mounted on the goniometer at this same angle during testing. Lamp circuit wiring, including consistent wire-to-lamp and wire-to-pin connections shall be maintained throughout the test, including during the bare-lamp calibration procedure.

6.1.1 Goniophotometer. A goniophotometer[3] is a partially or fully automated electromechanical system that positions a lamp and/or luminaire and measures the directional distribution of light. The goniophotometer and mounting apparatus should be rigid enough to prevent angular deflection of the lamp and/or luminaire and the goniophotometer structure, even when there is appreciable off-balance loading from the lamp and/or luminaire and its mounting apparatus. The goniophotometer and associated mounting apparatus should minimize interference with light emitted by the lamp and/or luminaire. Fluorescent lamps exhibit some luminous output change as a function of lamp orientation. Type C goniophotometers shall be used to avoid light output change due to lamp tilting with respect to gravity.

6.1.1.1 Photodetector. The photometer head, or photodetector,[4] of the goniophotometer should have relative spectral responsivity matched to the V(λ) function. The f_1' value of the spectral responsivity shall be less than 3%. For absolute measurements on spectrally structured sources (e.g., RGB or monochromatic), a systematic spectral correction factor should be considered. To determine such a correction factor, the spectral responsivity of the photometer head and the spectral distribution of the source to be measured should be known. It is important to note that there are photometer heads readily available that have an f_1' value of spectral responsivity less than 1.5%.

For determining total luminous flux of a luminaire or lamp using a goniophotometer, the photometer head shall have a good cosine response within the angle range where light is incident, with the $f_2(\varepsilon,\o)$ value (relative deviation from the cosine function) of less than 2% within the acceptance angle range. In this application, the acceptance angle range is defined as the span of angles across which the photodetector will be expected to measure during a goniophotometric exploration. In most cases, this angle range will be less than 20 degrees. This good cosine response is required in order to accurately measure the candela values needed to calculate total luminous flux.

6.1.1.2 Angular Positioning. The uncertainty in angular position due to all factors (e.g., encoder accuracy, structural deflection, mirror alignment, reversal error) should be no more than 0.1 times the required measurement increment.

6.1.1.3 Repeatability of Light Readings. One purpose of goniophotometry is to accurately measure the spatial distribution of light for a lamp or luminaire. The goniophotometer should have good repeatability in light readings. Many factors contribute to the uncertainty in light readings, and the goniophotometer should be evaluated to determine whether the system's measurement uncertainty is consistent with the measurement needs of the lamp or luminaire under test. Some factors influencing the system uncertainty are:

- Positioning repeatability
- Accuracy of gain ratios in amplifier and scaling circuits
- Positioning accuracy
- Quantization error in analog-to-digital (A/D) conversion
- Linearity in the A/D conversion
- Spectral responsivity of the photodetector vs. the source spectral properties

The uncertainty factors for any given goniophotometer depend on the design and construction of the instrument and the relative contributions of the factors appropriate for that system. As a general goal, light readings within a single test setup should have repeatability of within ±2%.

6.2 Test Distance

The test distance (see **Section 3.4** and **Figure 6-1**) shall be great enough that the inverse-square law[5] applies. A test distance greater than or equal to 5 times the largest dimension of the luminous opening(s) should be sufficient for most fluorescent roadway and area lighting luminaires. The luminaire optical properties should always be evaluated in regards to the minimum required distance. Test distances that are insufficient to satisfy the inverse-square law will cause errors in the reported luminous intensity values. In all cases, the test distance shall be reported. (See **Figure 6-1**.)

6.3 Selection and Preparation of Luminaire

6.3.1 Luminaire, Lamp and Ballast Selection. The luminaire and lamp shall be selected to meet the requirements of the test(s). Electrical details shall meet American National Standards Institute (ANSI) specifications.[6,7] Ballasts regularly furnished as part of or recommended for use with the luminaire should be used to operate the lamp(s) during the test, and shall be mounted in their normal locations. Ballasts should be within tolerances stated in the current ANSI specification[8] for fluorescent lamp ballasts.

6.3.2 Cleaning of Luminaire Optical Parts. All optical parts shall be thoroughly cleaned before measurements are made, unless the purpose of the test is to determine the effect of dirt on the luminaire. For the test that follows a dirt depreciation test, the luminaire shall be returned to as near the "like new" condition as possible. Use of corrosive or abrasive materials in cleaning could alter the finish of the lamp, reflector, refractor or glass cover, and shall not be used.

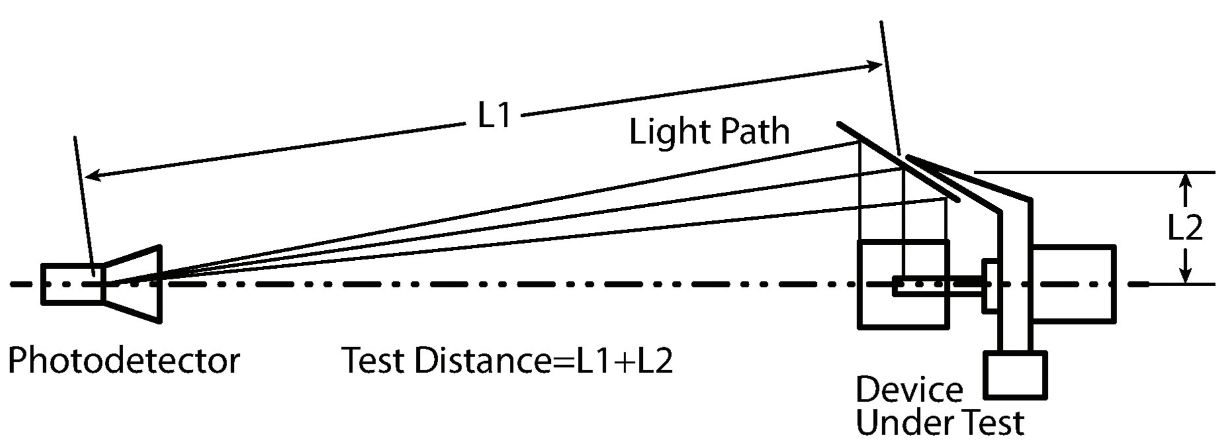


Figure 6-1. Diagram of test distance for a typical moving-mirror goniophotometer. (© Illuminating Engineering Society)

6.4 Test Lamps

6.4.1 General. Lamps with stable light output are needed for photometric test purposes, and their light output should remain stable during a test or test series when constant line voltage is supplied. Phosphor-coated lamp(s) should be carefully selected to obtain a phosphor density that is uniform over the entire lamp envelope. Lamps designed for a symmetric distribution should appear uniform in brightness and color from end to end and around the axis. If the intensity distribution of a lamp in a plane at 90 degrees to its long axis is designed to be other than circular, it should be checked to make certain that the shape of the intensity distribution is representative of its type.

6.4.2 Axially Symmetric Lamps. The circular distribution should be uniform within ±2% (a 4% spread). Readings should be taken in a minimum of four locations, 90 degrees apart, around the longitudinal lamp axis. It is preferable that data at finer increments be taken.

The test lamp should be mounted in a vertical or horizontal position, with a mounting that shall not shadow the lamp or that can be altered after mounting to remove any shadowing. Stray light corrections may be necessary to obtain net values. Uniformity of the intensity distribution is determined thus:

$$U = \frac{2(I_{max} - I_{min})}{I_{max} + I_{min}}$$, expressed as a percentage,

where:

U = Uniformity

I_{max} = Maximum reading observed during 360° scan

I_{min} = Minimum reading observed during 360° scan

If U exceeds 4%, the lamp should be rejected.

6.4.3 Matching Multiple Lamps in Luminaires. Lamps for luminaires designed for operation with more than one lamp should be matched for light output within ±1.5% (a spread of 3%) when operated on the same supply and ballast circuit.

6.4.4 Lamp Seasoning, Pre-burning and Stabilization. Lamp seasoning shall be conducted as described in ANSI/IES LM-54-20 (see **Section 2.3**). Once seasoned, all lamps should be carefully protected against bumps or jolts during handling.

Because the distribution of mercury within the discharge tube significantly affects the electrical and photometric characteristics of the lamp, pre-burning is necessary to obtain repeatable measurements.

For many types of fluorescent lamps, the time required for pre-burning is relatively long, and it may be necessary to pre-burn lamps at a location separate from the photometric equipment. In this case, the test lamp may be extinguished at the end of the pre-burning period and then transferred to the test position. As it is moved from one location to the other, it is important to keep the test lamp in the same physical orientation as maintained during pre-burning. Care shall be taken to avoid shaking or bumping the lamp during transfer, as this can cause mercury to dislodge from the cool zones. The lamp should be less sensitive to movement if it is allowed to cool down for 15 minutes before being transferred.

When a test lamp is pre-burned in a separate location, it may be necessary to operate the lamp under test conditions for some additional period of time to achieve a stable output. Lamp and ballast should reach stable operating conditions (particularly with respect to temperature) before testing. A false sense of stabilization can occur if the sampling period is short compared to the normal lamp stabilization time. Lamp stabilization readings should be taken at approximately 15-minute intervals. The lamp is considered stabilized when the light output over the last 30 minutes produces fluctuations no greater than ±0.5%. For 800-mA or 1,500-mA lamps, or for compact fluorescent lamps, stability is reached when at least 5 readings of the light output and electrical power over a period of 60 minutes, taken 15 minutes apart, is less than 0.5%. This assumes previous stabilization of lamps with controlled mercury pressure regions.[9] Successive light readings shall not be increasing or decreasing monotonically. Results under these conditions can be correlated with additional tests where the same luminaire is subjected to various thermal and air circulation environments considered to be typical of in use conditions. Lamps shall not be restarted during a test.

Recommendations for pre-burning and stabilization of specific types of fluorescent lamps are described below.

6.4.4.1 Compact Fluorescent Lamp (CFL) Considerations. The single ended CFL is a low-pressure mercury discharge light source, with compact geometry achieved by using two or more folded arc tubes or two or more individual tubes connected by a bridge. Some designs include a "cold chamber," which provides control of the mercury vapor pressure. Circline and U bent lamps are not designed with specific cold chambers and are not as dependent on operating position for proper mercury migration. Circline and U bent lamps do not require extensive pre-burning to reach stable output and may be pre-burned and stabilized in the test position.

Single-ended CFLs are generally more difficult to stabilize for measurements than conventional linear or circline lamps. CFLs, other than circline and U-bent lamps, should be pre-burned for 15 hours in a based-up position at within ±5% of rated voltage. Ambient temperature should not exceed 40 °C to ensure that excess mercury condenses in the coolest parts of the lamp.[10,11] If CFLs are seasoned in the base-up position and care is taken in the physical handling of the test lamp, the pre-burning requirement will be substantially shortened, and the lamp may be stabilized in the test position.

The electrical and photometric properties of CFLs are highly dependent on the mercury vapor pressure within the discharge tube, unless it is one of the specialized amalgam types. This dependency is affected by the location of the mercury within the lamp.

Note: The peak light output of some CFLs may not be achieved at ambient temperatures of 25 °C (77 °F).[10,11] Some single ended CFLs are designed with cold chambers consisting of tips that extend away from the path of the discharge. This provides a cooler place for the mercury to condense, lowering the mercury vapor pressure within the lamp and thus raising the ambient temperature value at which the peak light output occurs.[10,11]

6.4.4.2 Considerations for Compact Fluorescent Lamps (CFLs) in Non-base Up Positions. There are widespread applications where CFLs are operated base down or horizontally. In the base down position, the cooling tips or cold chambers will be located at the upper most part of the lamp. As the lamp continues to operate in this orientation, mercury will migrate to the cooling tips just as it would if the lamp were oriented base up. However, in a base down orientation, minor vibration or movement of the lamp can cause mercury droplets to fall to the bottom or cathode end of the discharge tube, thus raising the mercury vapor pressure and causing the luminous flux of the lamp to change. Lamps tested base down will therefore tend to exhibit random cyclical variations of light output and electrical values.

For lamps having more than two discharge tubes, there will usually be cooling tips at both ends of the lamp. In the base down position, the mercury will collect at the tips in the lowest part of the lamp, which are coolest. Therefore, stability can be achieved in the base down position, as well as the base up position, and the test lamp should be handled according to the recommendations of this document. For base down photometry, lamps shall also be pre-burned and stabilized base down.

For lamps operated in the horizontal position, stability can be achieved, but a much longer time may be required. The recommended standard method shall be that the lamp first be pre-burned base up until stabilization can be achieved. The test lamp can then be moved slowly without vibration to a horizontal position and re-stabilized. Measurements can then be taken when a stable luminous flux value is reached.

6.4.4.3 Twin-Tube Fluorescent Lamp Considerations. Twin-tube fluorescent lamps are typically seasoned in a horizontal position such that a plane passing through the central axis of both tubes shall be parallel to the floor. After seasoning, the lamp should be maintained in the same horizontal position, so as to minimize the time required to pre-burn and stabilize output.

Twin-tube fluorescent lamps shall be mounted in the luminaire in the same orientation in which they will be pre-burned and should be pre-burned for a minimum of 12 hours to ensure thermal stabilization.

The lamp should be maintained in the horizontal position when transporting and when inserting it into the luminaire. At all times, the base of the lamp should be level with or higher than the opposite bend area of the lamp so that the mercury will remain in the cold spot location that has been established in the bend area. This procedure should be carefully followed so that significant errors are not induced in measuring light output.

When inserting the lamp into the luminaire, the horizontal position should be maintained as dictated by the specific luminaire lamp holder geometry.

6.5 Test Methodologies

6.5.1 General. The lamp(s) or luminaire shall be operated under conditions as close to typical intended end-use as possible, or suitable corrections made. For example, if a luminaire is to be mounted on a pole at a tilted angle, it shall be mounted on the goniophotometer at this same angle.

6.5.2 Bare Lamp Measurement Conditions. When measuring the test lamp(s), the following conditions should be observed:

1. The lamp(s) shall be mounted and operated in the same orientation as they would be if they were in the luminaire. The light center of the test lamp (or the geometric center of the light centers of the test lamps) shall be mounted at the goniophotometer center.
2. The instruments (i.e., the photodetector and electrical measuring instruments) and source parameters (i.e., the test distance and the line voltage) should be the same as used for the luminaire.
3. The lamps shall be operated on the same circuits using the same ballasts and the same pin polarities as used for the actual test. All lamps that operate from the ballast shall be operated during lamp measurement, and at the same ambient temperature, even if the photodetector measures the output of each lamp separately. Inadvertent modification of the bulb wall temperature should be avoided. If lamps are measured in a group, they shall be mounted to prevent inter-reflections between lamps and mutual heating effects (e.g., use baffles or lamp orientation to prevent this). Care should be taken that any baffles used do not cause heat buildup. If multiple single bare-lamp measurements are required, the total bare-lamp reading is the sum of the individual readings.
4. Lamps and ballasts shall reach stability as defined in **Section 6.4.4** before photometric readings are taken. Electronic monitoring should be used to determine when electrical and light readings become stable.
5. If the bare-lamp measurement immediately follows a luminaire distribution test and the luminaire is already stable, less time may be required to reach a stable operating condition with the bare lamp(s). This requires moving the lamps from the thermally stable luminaire to the bare-lamp rack and reenergizing the lamps as quickly as possible. Care should be taken to prevent tilting or jarring the lamp to avoid mercury redistribution.
6. Lamp lumen ratings should be taken from the lamp manufacturer's current listing for the particular lamp used in the test. If the lumen value is not for an ambient temperature of 25 °C (77 °F), the lamp manufacturer should be consulted for lamp testing guidelines.

6.5.3 Lamp Photometric Calibration Process. There are two basic methods to generate goniophotometric test data: relative photometry and absolute photometry. Each method has advantages and disadvantages.

In relative photometry, the goniophotometer is used to measure the test lamp independent of the luminaire, to create a value typically referred to as the bare lamp sum. The *bare lamp sum* is a number proportional to the lumen output of the lamp. This bare lamp sum is then used to normalize the intensity data of the luminaire. The normalized luminaire data can then be rescaled by the manufacturer's initial rated lumen output for the test lamp to create report data, or "IES file" data, that are prorated for the lamp's rated lumen output. From a practical standpoint, the relative method is desirable because a traceable photometric standard is not necessary.

In absolute photometry, the goniophotometer's responsivity in candelas or counts is established using a mathematical model and system component data (e.g., mirror reflectivity, photodetector responsivity), or by measuring the system response to a traceable standard of luminous intensity or luminous flux. The calibrated goniophotometer can then be used to measure the properties of a light source or a luminaire in terms of absolute luminous intensity. Absolute photometry is advantageous or necessary when light source properties in the luminaire are strongly influenced by thermal effects such that the sources cannot be independently measured. The drawback of absolute photometry is that it requires the maintenance of a traceable standard of luminous intensity.

6.5.3.1 Relative Photometry Method. The relative method consists of the following steps:

1. Measure the test lamp (i.e., the "bare lamp").
   - Measure the test lamp's luminous intensity distribution in the same position, under the same electrical operating conditions, and on the same goniophotometer as will be used for the luminaire test.
   - Use a measurement angle set appropriate for the lamp position. (Refer to the subsections within **Section 6.5.3.1** for appropriate angle sets and methods.)
2. Measure the luminaire.
   - Measure the luminaire's luminous intensity distribution using the same test lamp, under the same electrical operating conditions, and on the same goniophotometer as was used for the bare lamp test.
   - Use a measurement angle set appropriate for the luminaire's expected photometric distribution type. (Refer to the subsections within **Section 6.5.4.4** for appropriate angle sets.)
3. Process the luminous intensity data.
   - Determine the ratio (R_G) of the bare lamp test signal gain to the luminaire test signal gain.
   - Determine the manufacturer's initial lamp lumen rating for the test lamp.
   - Compute the test lamp's bare lamp sum by summing the product of the measured intensity data and the zonal constants for the measurement angle set. (Refer to **Annex A** for information on computing zonal constants.)
   - Scale the measured intensity data to create corrected luminous intensity values that are prorated to the test lamp's rated initial lumen output:

$$I_c = \frac{I_m \times LRR}{BLS} \times R_G ,$$

where:

I_c = corrected luminous intensity value of luminaire at a given angle

I_m = measured luminous intensity value of luminaire at a given angle

BLS = bare lamp sum

LLR = lamp lumen rating

R_G = ratio of measurement gains for the bare-lamp and luminaire tests

4. Compute the luminaire efficiency.
   - Compute the luminaire sum by summing the product of the measured luminaire intensity data and the zonal constants for the luminaire's measurement angle set. (Refer to **Annex A** for information on computing zonal constants.)
   - Compute the efficiency:

$$E_L = \frac{LS}{BLS} \times R_G ,$$

where:

EL = luminaire efficiency

LS = luminaire sum

BLS = bare lamp sum

R_G = ratio of measurement gains for the bare lamp and luminaire tests

It is not necessary to measure the bare lamp before the luminaire. For linear fluorescent lamps, an alternative method can be used to measure and compute the bare lamp sum, as described in **Section 6.5.3.1.3**.

6.5.3.1.1 Vertically Mounted Compact Fluorescent Lamps. Most single ended compact fluorescent lamps require that all excess mercury be condensed in the cold spots for proper stabilization. Compact fluorescent lamps shall not have their position altered during photometry. These lamps may be rotated about the vertical axis during photometry, but lamp-operating position shall not be altered.

Luminous output for vertically mounted compact fluorescent lamps shall be determined by measurements in at least 16 lateral half-planes spaced equally about the longitudinal lamp axis.

The vertical angle increments within the half-planes shall be sufficiently dense to represent the luminous intensity distribution such that each data point is a good approximation to the average luminous intensity within its zone. For 10-degree zones, vertical angle measurements shall be at 0, 5, 15, 25, 35 … 165, 175, and 180 degrees. If necessary, finer angular increments may be used to properly represent the intensity distribution.

6.5.3.1.2 Horizontally Mounted Compact Fluorescent Lamps. Horizontally mounted test lamps shall be measured in their end-use position. The lamp shall be positioned about its longitudinal axes in a manner identical to that used for the luminaire. Horizontal lamp measurements shall be performed in an adequate number of vertical half-planes; a minimum of 16 lateral half-planes spaced equally shall be employed to accurately describe the luminous intensity distribution. Care should be taken to avoid measuring in planes that may be shadowed by the goniophotometer support system or other structural items (e.g., the 0° or 180° plane would typically contain a support structure for the lamp socket).

The vertical angle increments within the half-planes shall be sufficiently dense to represent the luminous intensity distribution so that each data point is a good approximation to the average luminous intensity within its zone. For 10-degree zones, vertical angle measurements shall be at 0, 5, 15, 25, 35 … 165, 175, and 180 degrees. If necessary, finer angular increments may be used to properly represent the intensity distribution.

6.5.3.1.3 Linear Lamp K-Factor. The measured relative lamp lumens for a linear fluorescent lamp with a symmetric light distribution may be determined by measuring the average intensity perpendicular to the lamp and applying a factor, *K*. *K* is the ratio of lumens to normal (i.e., perpendicular) luminous intensity, such that:

$$relative\ lamp\ lumens = K \times (90^\circ\ relative\ luminous\ intensity),$$

where 90° *luminous intensity* refers to the intensity perpendicular to the lamp axis and through the centroid of the lamp.

Relative lamp lumens are then determined by multiplying the 90° intensity value by a known *K*-factor.

K is given by:

$$K = \frac{9.15}{\frac{D}{L}\operatorname{atan}\left(\frac{L}{2D}\right) + \frac{2D^2}{4D^2 + L^2}},$$

where:

K = lumen/luminous intensity ratio

D = test distance in meters

L = lamp length in meters

atan is in radians

Note: This ratio relates solely to the light distribution of the lamp and therefore is constant regardless of the setting of the photometer measuring circuit.

The above equation for K is shown in **Figure 6-2** as a family of curves for lamps of 0.6, 1.2, 1.8, and 2.4 m [2, 4, 6, and 8 ft] in length and test distances from 3 to 18 m [about 10 to 60 ft]. To arrive at the appropriate *K* factor, the intersection between the length of the lamp under test and the test distance used is found, and the *K* value is read on the left edge of **Figure 6-2**.

6.5.3.1.4 Extra High Output Non-symmetric Linear Lamp K-Factor. The light output of extra high output (EHO) lamps is sensitive to orientation, ambient temperature fluctuations, and drafts, as described in **Annex B**. The determination of calibration luminous intensity, normal to the longitudinal axis of the lamp,

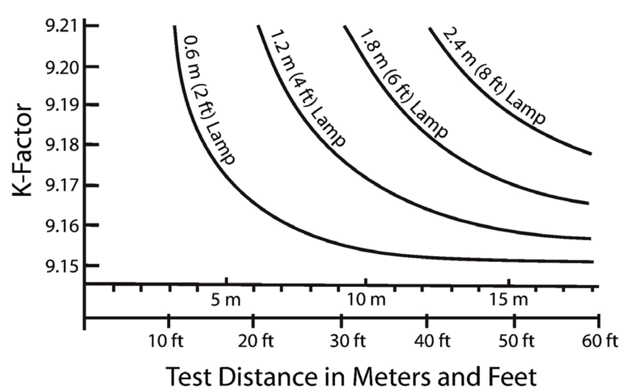


Figure 6-2. A plot of the K-Factor with axial symmetry of various lengths and test distances. (© Illuminating Engineering Society)

shall be determined without moving any of these lamp types. Inversion to allow viewing of the top and bottom of these lamps from below, for example, will give erroneous results.

Frequently, the photometer construction will be such that it is not possible to view the entire lamp distribution without obstruction. In these cases extrapolation shall be used to fill in that portion of the lamp distribution which cannot be measured without obstruction. The region over which such extrapolation is applied should be kept to the minimum possible. In no case should the region of extrapolation exceed 30 degrees of cone (for example, extrapolation of the region 150 to 180 degrees is acceptable; extrapolation of the region 140 to 180 degrees is not acceptable).

6.5.3.1.5 Nonlinear Lamp K-Factor. Non-linear lamps (such as circline and U-shaped) can be calibrated in terms of relative lumen output in the same manner as described for compact fluorescent lamps per **Sections 6.5.3.1.1** and **6.5.3.1.2**. If the lumen-to-luminous-intensity ratio is determined by complete photometry, it is important to maintain the highest possible accuracy. Errors (random and systematic) in calibration will directly add to the limits of accuracy inherent in the luminaire photometry.

Alternatively, a *K*-factor can be measured for these lamps. Separate determination of a *K*-factor shall be performed for each lamp type, as the formula provided in **Section 6.5.3.1.3** for linear lamps is inapplicable for non-linear types because of lamp inter-reflections.

To determine *K*-factor for a non-linear lamp, relative lamp lumens are determined by the method described for compact fluorescent lamps. From the test lamp data, the 90° relative luminous intensity value is determined. The K-factor is then calculated from the ratio of relative lamp lumens to the 90° relative luminous intensity value. The *K*-factor so determined can be used for calibration of other lamps of identical geometry and light distribution.

It is important to note that non-linear lamps are generally orientation sensitive, and the precautions given in **Section 6.5.3.1.4** apply.

6.5.3.2 Absolute Photometry Method. There are two methods for conducting absolute photometry of roadway and area lighting fluorescent luminaires. In the first method, the goniophotometer detector system is calibrated using a standard of luminous intensity or a standard of luminous flux (standard lamp). The second method is based on the use of a calibrated photometer.

6.5.3.2.1 Method Using Luminous Intensity Standard. A standard of luminous intensity is a light source that has been calibrated by a laboratory capable of establishing an absolute value of luminous intensity. A calibrated luminous intensity standard is mounted in the goniophotometer such that measurements can be made in the specified orientation and direction for which the luminous intensity standard is calibrated. The calibrated luminous intensity standard is powered in accordance with the instructions provided by the calibration laboratory, and measurements are made using the goniophotometer photodetector along the normal light (i.e., perpendicular) path inclusive of any system mirrors or other optical elements along the light path.

It is recommended that three standards be used to reduce the calibration uncertainty. The ratio of the calibrated luminous intensity values to the measured values of the calibrated standards are averaged to establish a calibration factor for the goniophotometer.

An example calculation is provided using three luminous intensity standards:

Luminous Intensity Standards	Photometer Readings
Standard 1: 1,500 cd	Standard 1: 3,782 counts
Standard 2: 1,550 cd	Standard 2: 3,908 counts
Standard 3: 1,532 cd	Standard 3: 3,862 counts

Calibration Factor

Cal Factor 1 = 1,500/3,782 = 0.3966

Cal Factor 2 = 1,550/3,908 = 0.3966

Cal Factor 3 = 1,532/3,862 = 0.3966

Cal Factor Avg. = 0.3966 cd/count (at measurement gain settings)

This calibration factor is good for the system gain setting at which the calibration measurements were made. For other system gain settings, the factor is scaled in proportion to the gain ratios. Goniophotometers require a wide dynamic range to handle the range of candela values generated from bare lamps to beam-forming luminaires. Programmable amplifiers and high-resolution analog-to-digital converters normally provide this dynamic range. To establish an absolute system sensitivity that is independent of the gain settings, knowledge of the electrical signal processing is applied to compute the candelas per ampere of photocurrent. **Figure 6-3** shows a block diagram of typical signal processing in a goniophotometer.

6.5.3.2.2 Method Using Luminous Flux Standard. A standard of luminous flux is a light source that has been calibrated by a laboratory capable of establishing an absolute value of luminous flux. A calibrated luminous flux standard is mounted in the goniophotometer such that measurements can be made in the specified orientation for which the luminous flux standard is calibrated (i.e., typically vertical base up or vertical base down). The calibrated luminous flux standard is powered in accordance with the instructions provided by the calibration laboratory, and measurements are made using the goniophotometer photodetector along the system light path inclusive of any system mirrors or other optical elements along the light path.

It is recommended that three standards be used to reduce the calibration uncertainty. The ratios of the calibrated luminous flux values to the measured values of the calibrated standards are averaged to establish a calibration factor for the goniophotometer.

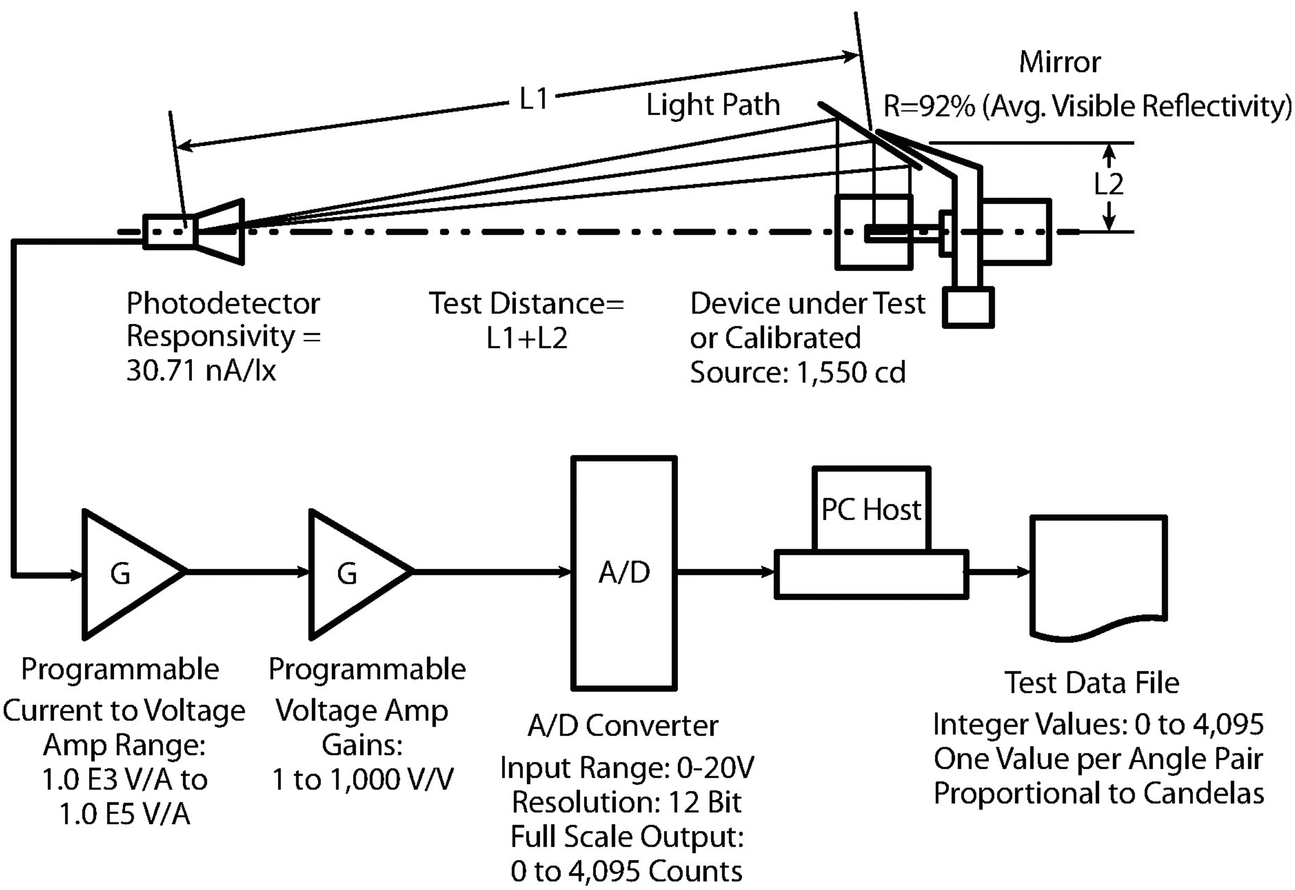


Figure 6-3. A block diagram of typical signal processing in a goniophotometer. (© Illuminating Engineering Society)

An example calculation is provided using three luminous flux standards:

Luminous Flux Standards	Photometer Readings
Standard 1: 5,420 lm	Standard 1: 13,670 counts
Standard 2: 5,456 lm	Standard 2: 13,756 counts
Standard 3: 5,382 lm	Standard 3: 13,572 counts

In this example, the counts are representative of the "lumen sum" of the luminous flux standard. In other words, the counts are representative of the sum of the products of the counts (proportional to the candelas) at each measurement angle and the zonal constants for the angular range of that measurement angle at the calibration gain setting.

Calibration Factor

Cal Factor 1 = 5,420/13,670 = 0.3965

Cal Factor 2 = 5,456/13,756 = 0.3966

Cal Factor 3 = 5,382/13,572 = 0.3966

Cal Factor Avg.: 0.3966 lm/(count·sr) = 0.3966 cd/count (at the calibration gain setting)

Note: cd = lm/sr

This calibration factor is good for the system gain setting at which the calibration measurements are made. For other system gain settings, the factor is scaled in proportion to the gain ratios or calibrated directly at additional gain settings.

This system calibration factor (Sys Cal Factor) represents the system's sensitivity inclusive of all optical factors such as path length, mirror reflectivity, and photodetector responsivity specific to the goniophotometer in use and can be scaled as necessary to account for different signal processing settings, such as amplifier gains.

6.5.3.2.3 Method Using Calibrated Photometer. Absolute photometry using a calibrated photometer[12] may be conducted by using the calibrated photometer as the detector system of the goniophotometer, or by transferring the calibration to the goniophotometer photodetector. The photometer's calibration should be valid over the full range of measurement values encountered during testing of the bare lamp and the luminaire; therefore, aspects such as detector linearity and spectral mismatch should be considered and preferably corrected.

The goniophotometer system calibration is determined by using the calibrated photodetector and the reflectance of any system mirrors or transmission of other optical elements along the light path under the intended conditions of use. For example, if the goniophotometer being calibrated uses a back surface mirror with source and detector aligned at 42° from the mirror surface normal, then the mirror reflectance, inclusive of the glass transmission losses, should be known at that angle of incidence. This system calibration is good for the determination of total luminous flux of a bare lamp or a luminaire. To determine the candela distribution of a source, the distance between the detector reference plane and the light source reference plane should be known. An example calculation is provided for both system calibrations (one for total luminous flux and one for luminous intensity).

Example of system calibration in lx/count using a calibrated photometer:

A national metrology institute (NMI)-traceable calibrated photometer corrected to the V(λ) CIE photopic standard observer with f_1' of less than 3% has:

Abs Cal = 88.67 nA/lx = 8.867×10^{-8} A/lx

Abs Cal = 1.128×10^{7} lx/A

Gain Scaling Factor:

Preamp Transimpedance gain (G1): 1.0×10^{5} V/A

Post-amp Gain (G2): 100 V/V

Gain Factor = G1 × G2 = 1.0×10^{7} V/A = 1.0×10^{-7} A/V

Analog-to-Digital Converter Parameters:

12-Bit A/D Converter: 4,096 counts, full scale

Input Signal Range: ±10 V, full scale

A/D Factor: 4.8828125 mV/count

A/D Factor: 204.8 counts/V

Mirror Reflectance (at angle of use):

Reflectance = 0.92 (for CIE illuminant A)

It is important that the mirror reflectance be representative of the spectral and geometric conditions of use in the goniophotometric measurements.

Calibration Factor (for this gain setting):

Cal Factor = Abs Cal × Gain Factor × A/D Factor/ Mirror Reflectance

Cal Factor = $(1.128 \times 10^{7}$ lx/A)$(10^{-7}$ A/V)$(4.883 \times 10^{-3}$ V/count)/0.92

Cal Factor = 5.987×10^{-3} lx/count

Example of system calibration in cd/count using a calibrated photometer:

Candela Calibration Factor (for this gain setting):

Cd Cal Factor = [Cal Factor (refer to previous example)] × [Distance (L1 + L2)]^2

Cd Cal Factor = $(5.987 \times 10^{-3}$ lx/count) $(6.12\text{m} + 1.22\text{m})^2$

Cd Cal Factor = 0.3967 cd/count

For absolute photometry, the calibration of the standard lamp or the standard photometer shall be traceable to a national metrology institute (NMI). In addition, the laboratory should determine the measurement uncertainty when transferring standard values.

6.5.4 Luminaire Photometry. The following sections describe the physical setup for performing the photometric tests.[13]

6.5.4.1 Mounting. The luminaire should be mounted on a goniophotometer designed for minimum obstruction of light. Care should be exercised to prevent light from reflecting off of the mounting apparatus directly into the photodetector or back onto the luminaire where it may artificially increase luminaire intensities.

6.5.4.2 Luminaire Position Sensitivity. Fluorescent luminaires can be sensitive to the effects of mounting orientation. Even the most stable linear type fluorescent lamps can exhibit differences in light output characteristics as a function of orientation.

Therefore, luminaires shall be tested only in their intended mounting orientation.

This precaution avoids test-related mechanical disturbances such as sag, warping, and shifts in the position of critical light controlling components; temperature variations due to changes in airflow; and changes in lamp intensity distribution due to orientation changes.

6.5.4.3 Luminaire Positioning on a Goniophotometer. Luminaires shall be positioned correctly relative to the goniophotometer axes to correctly measure the luminous intensity distribution. Luminaires shall be positioned on the goniophotometer according to the following guidelines:

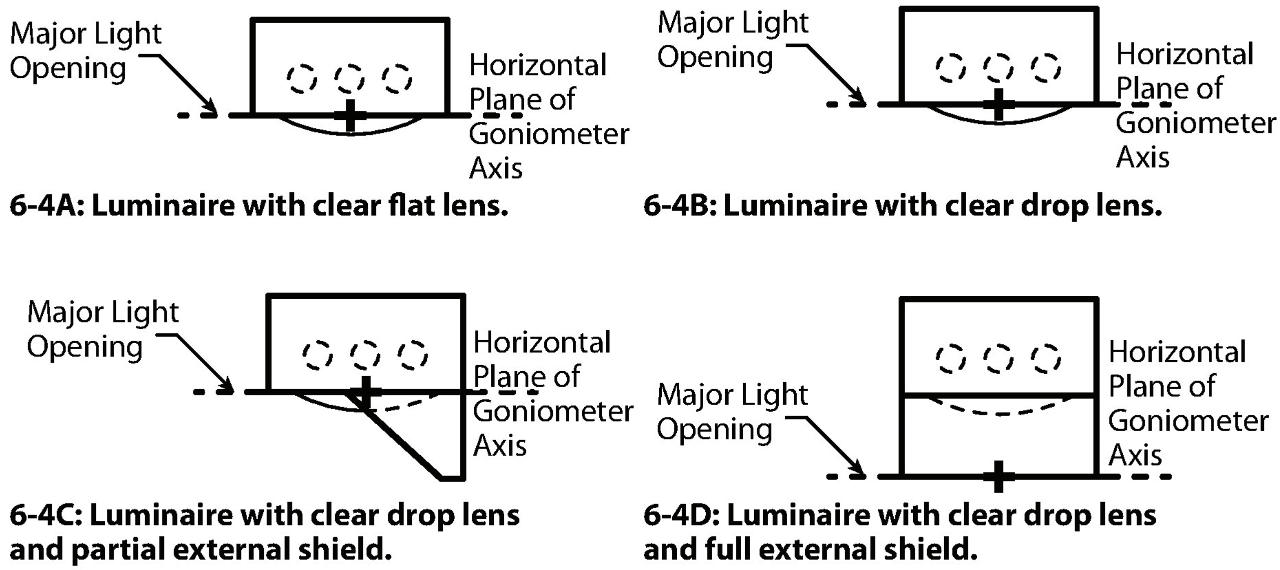
1. If the light center of the test lamp is above the reflector opening, or above the plane of the major light center of the reflector opening, the luminaire shall be mounted so that the plane of the major light-emitting opening coincides with the goniophotometer center. Examples of luminaires meeting this criterion and the location of the goniophotometer center are shown in **Figures 6-4** and **6-5**.
2. If the light center of the test lamp is clearly visible, and outside of any reflector, the luminaire shall be mounted on the goniophotometer so that the light center of the lamp is at the goniophotometer center. (See **Figure 6-6A**.)
3. If a translucent aperture obscures the light center of the test lamp, the luminaire shall be mounted so that the central plane of this aperture coincides with the goniophotometer center. (See **Figure 6-6B**.)

Note: For the above examples, if the luminaire uses more than one lamp, the geometric center of all lamp light centers is considered in place of the light center of any individual lamp.

6.5.4.4 Selection of Angles for Photometric Measurements.

6.5.4.4.1 General. Angular relationships shall be:

- **Vertical:** Zero degrees is at nadir, 90° at horizontal, and 180° at zenith.
- **Lateral:** Angles are measured counterclockwise when viewing the luminaire from above, with 0°orthogonal to the curb toward the street side; 90°parallel to a curb line; 180° orthogonal to the curb toward the "house side"; and 270° parallel to a curb line.



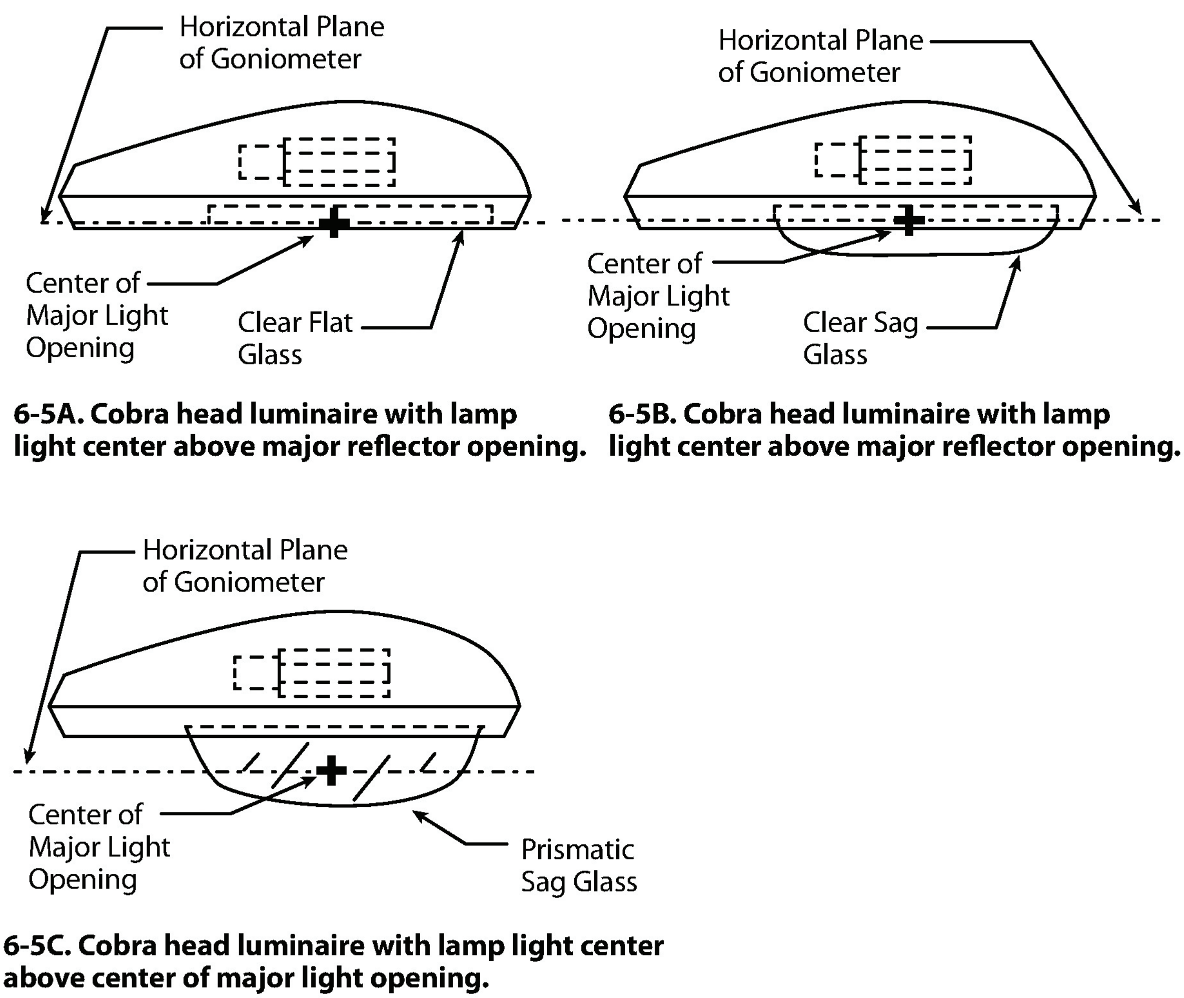
6-4A: Luminaire with clear flat lens.

6-4B: Luminaire with clear drop lens.

6-4C: Luminaire with clear drop lens and partial external shield.

6-4D: Luminaire with clear drop lens and full external shield.

Figure 6-4. Examples of proper luminaire and goniophotometer positioning. The geometric arc center of the lamps is above the major light opening. (© Illuminating Engineering Society)



6-5A. Cobra head luminaire with lamp light center above major reflector opening.

6-5B. Cobra head luminaire with lamp light center above major reflector opening.

6-5C. Cobra head luminaire with lamp light center above center of major light opening.

Figure 6-5. Cobra head luminaires illustrating various lamp light center cases. (© Illuminating Engineering Society)

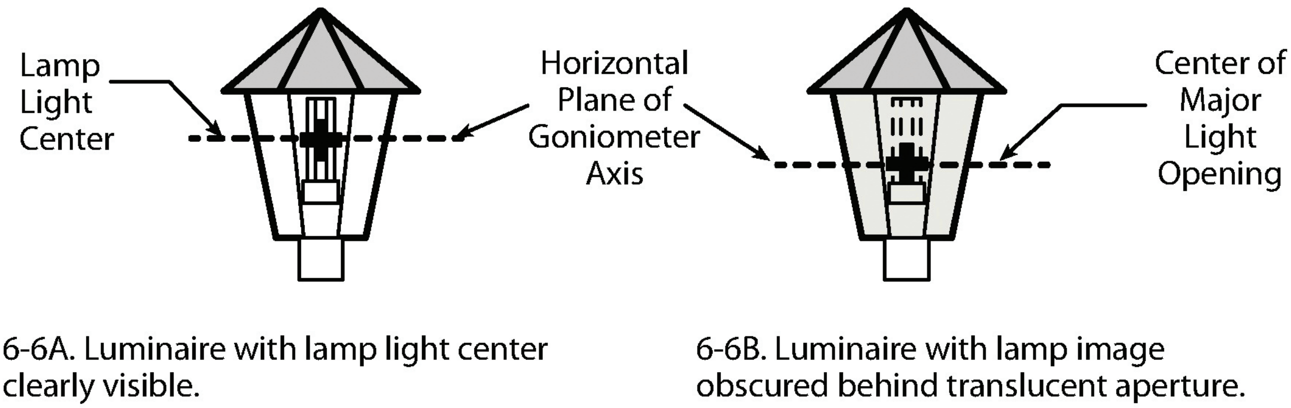


Figure 6-6. Examples of the proper luminaire and goniophotometer positioning. The light center is fully visible (A) or partially obscured (B). (© Illuminating Engineering Society)

Sufficient data shall be taken to accurately define the luminaire's luminous intensity distribution. The number of horizontal half-planes and the number of vertical angles at which readings should be taken depends on the horizontal and vertical rates of change in the luminaire's luminous intensity distribution. Rapid changes in luminous intensity can occur near the peak of a distribution and can significantly affect calculated veiling luminance values. As readings approach the angle of maximum luminous intensity, the separation of readings may need to be reduced to ensure that the distribution is accurately represented. Readings should be taken in 5-degree increments for vertical luminous intensity traces, but may need to be one degree or finer in regions with high rates of luminous intensity change. Increments less than five degrees may be desired to meet the recommendations of other needs, such as computing BUG ratings according to the definitions in TM-15-20.[14] For evaluation of a luminaire's BUG classification, the reading shall be made from 0 to 180 degrees vertical.

Similarly, readings should be taken in no greater than 10-degree increments between horizontal half-planes, and may need to be reduced in the region of the horizontal maximum. Use of smaller reading increments should be considered when luminous intensity changes exceed 20% between readings.

During a typical goniophotometric scan, the nadir (0 degree vertical) and zenith (180 degree vertical) angles are typically measured multiple times (once for each measured lateral plane). These multiple measurements are repeat measurements of the same angular location, and therefore the nadir values should be averaged and reported for each plane. Similarly, the zenith values should be averaged and reported for each plane.

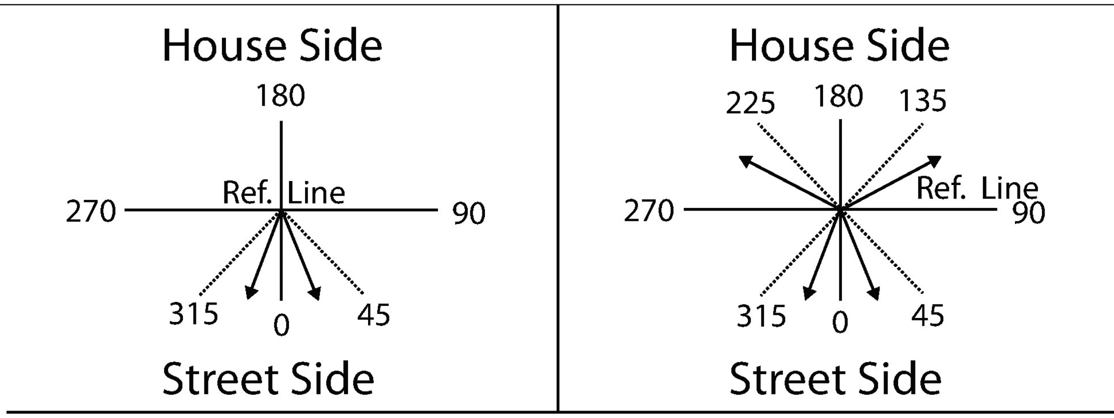
The point of maximum luminous intensity shall be located and reported.

6.5.4.4.2 Luminaires with IES Types I, I Four-Way, II, II Four-Way, III, IV, and VS Distribution. For luminaires having distributions that are symmetric with respect to a single vertical plane, readings shall be taken in vertical planes that are a maximum of ten degrees apart. Additional planes of data may be necessary where the distribution is changing rapidly.

Averages may be made of the values taken at corresponding angles on opposite sides of the plane of symmetry, as shown in **Figures 6-7**, **6-8** and **6-9**. Because of the averaging, it may be advantageous to laterally divide the distribution into ten-degree zones and measure at the mid-zone angles. Photometric parameters, such as the location of the maximum candela value, may then be measured on one side of the plane of symmetry.

6.5.4.4.3 Luminaires with IES Type V Distribution. For luminaires having a distribution that is symmetric about the vertical axis, readings shall be taken in a minimum density of ten-degree lateral increments and then averaged.

6.5.4.4.4 Luminaires with Asymmetric Distribution. For luminaires having asymmetric distributions, readings

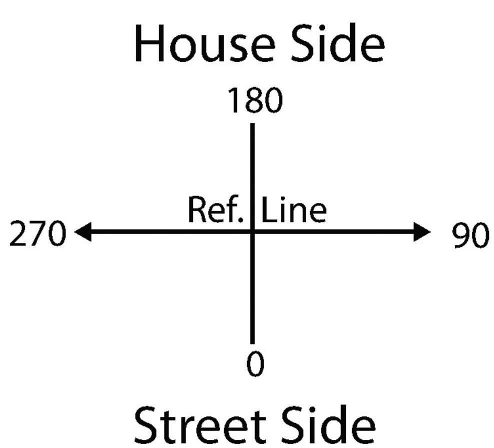


Minimum angles for photometric measurements for units with one plane of symmetry (0°-180°). Readings may be taken in vertical planes at the following lateral angles, for a total of 21 luminous intensity (candlepower) distribution curves (19 averaged and two direct readings):

0	90, 270	155, 205
5, 355	92, 265	165, 195
15, 345	105, 255	175, 185
25, 335	115, 245	180
35, 325	125, 235	
45, 315	135, 225	
55, 305	145, 215	
	165, 195	

A 45-degree shift of the reference line is made when reporting data for the 4-way distribution; either dotted line may be either parallel or perpendicular to the curb line. Arrows indicates lateral locations of main beams.

Figure 6-7. Averaging for distributions exhibiting bilateral symmetry (about the 0° – 180° lateral plane), such as Types II, III and IV. (© Illuminating Engineering Society)



0 - 180
5, 165, 185, 355
15, 165, 195, 345
25, 155, 205, 335
35, 145, 215, 325
45, 135, 225, 315
55, 125, 235, 305
65, 115, 245, 295

Figure 6-8. Averaging for distributions exhibiting four-quadrant symmetry (about the lateral planes 0 – 180 degrees and 90 – 270 degrees), such as Type I. (© Illuminating Engineering Society)

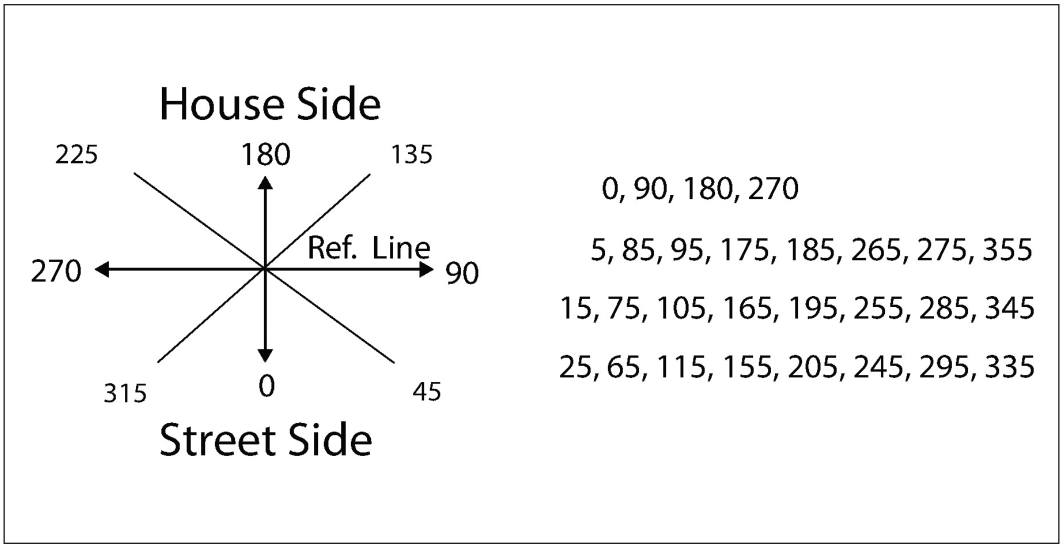


Figure 6-9. Averaging for distributions exhibiting octant symmetry (about the lateral planes 0 – 180, 45 – 225, 90 – 270, and 135 – 315 degrees), such as Type VS. (© Illuminating Engineering Society)

shall be taken in lateral half-planes and shall be a maximum of ten degrees apart. It may be necessary to take additional planes of data in the region of the main beam or where the distribution is changing rapidly. Since there is no symmetry, any computations performed shall be computed without averaging.

6.5.4.4.5 Area Luminaires. Some luminaires (such as outdoor wall packs, bollards, and general area lighting luminaires) may not be designed to fit into the standard Type I, II, III, IV, or VS classifications, and their light distribution may not be completely symmetric (as is the case in Type V luminaires, as described in **Section 6.5.4.4.3**). In these situations, the data shall be acquired in ten-degree or smaller lateral increments, and the luminaire should be classified as an area luminaire. If the light distribution of the luminaire exhibits bilateral, four-quadrant, or octant symmetry, appropriate averaging as described in **Section 6.5.4.4.2** may be applied.

6.5.5 Photometric Data Processing

6.5.5.1 Uplight Detection Threshold. Even with the most rigorous application of the techniques described in **Section 4.4** of this document, stray light will still contribute to the measured values in a photometric test. This can cause non-zero values to be measured at angles where the luminaire is not emitting light. These non-zero values are particularly problematic when the test results are intended to verify compliance with specifications that require zero light at certain angles (notably, the BUG ratings of the Lighting Classification System, LCS, described in IES TM-15-11[14]).

For example, the luminaires depicted in **Figures 6-4A** and **6-4D** (see **Section 6.5.4.3**) are designed with a horizontal flat glass enclosure with no external light directing devices. The design of this type of luminaire makes it impossible for light to be directed at or above 90 degrees vertical (90V) when the luminaire is mounted with the luminous opening horizontal. Because of the presence of stray light in the goniophotometer system, non-zero readings may still be recorded at these angles. If these spurious readings were reported as non-zero luminous intensity values, it would be impossible to classify the luminaire as "U0" according to TM-15-11.[14]

In order to avoid such misclassification of a luminaire, it is necessary to define the applicable test in terms of *detection*, rather than *measurement* of uplight. While it is impossible to verify that the uplight from any luminaire is exactly zero, it is possible to verify that a luminaire emits no uplight detectable by a particular measurement system. This necessarily involves determining the detection threshold of the measurement system and the comparison of the apparent uplight measured for the luminaire with that threshold.

For a luminaire designed to achieve zero uplight that is submitted for testing to support a U0 classification, non-zero values that exist in the test data at and above 90V may be ignored or assigned a zero value only if analysis of the data collected at and above 90V is consistent with the design-based assumption that the device under test produces zero uplight. This is the case when the apparent uplight is below the detection threshold of the measurement system, and attributable to the effects of stray light and noise in the detector signal.

Determination of the uplight detection threshold of a particular measurement system involves consideration of various system characteristics, including stray light and photometric signal-to-noise performance. In order to ensure reproducibility of test results, a *conventional goniophotometer system* is defined as one that has an uplight detection threshold equal to 0.25% of the total lumen output of the luminaire. Any measurement system used to perform U0 verification testing shall be shown to have an uplight detection threshold less than or equal to that of the conventional goniophotometer system. For the purpose of this standard, *detectable uplight* is defined as uplight detectable by the conventional goniophotometer system.

For a luminaire submitted for BUG testing, if the apparent uplight measured falls below 0.25% of the total lumen output of the luminaire, then luminous intensity values for angles at and above 90V may be omitted from the test report or assigned a zero value. The original data shall be retained for reference, and the details of the supporting detection threshold analysis shall be documented. In this case, the test report shall include the statement:

> *"The device under test emits no detectable uplight, as defined by ANSI/IES LM-10-20."*

If uplight values are assigned a zero value, the test report shall also include the statement:

> *"For the purpose of this report, certain non-zero uplight readings, attributable to instrument artifacts, have been assigned a zero value."*

6.5.5.2 Isoilluminance Diagram. An isoilluminance diagram of horizontal illuminance shall be shown on a horizontal plane. This plane shall have a width of at least 4 times the assumed mounting height (4 MH) on the street side of the reference line, 2 MH on the house side, and a length of 5 MH parallel to the reference line. (See **Annex C** for a discussion of the appropriate mathematical calculations.) It is recommended that enough isolines be shown so as to minimize the amount of interpolation needed. A numbering system such as 0.1 MH, 0.2 MH, 0.5 MH, 1.0 MH, 2.0 MH, 5.0 MH has been traditionally used, but a logarithmic system such as 0.1 MH, 0.2 MH, 0.4 MH, 0.7 MH, 1.0 MH, 2.0 MH, 7.0 MH often yields a more uniform pattern.

6.5.5.3 Isocandela Diagram. To classify the light distribution of the roadway luminaire in accordance with the ANSI/IES RP-8-18,[2] at least a maximum candela point and a half-maximum candela trace are required. These intensity traces may be included as part of the isoilluminance diagram.

6.5.5.4 Utilization Efficiency. A pair of curves representing street- and house-side utilization of the emitted flux shall be plotted. These curves indicate the percentage of flux falling on the area (of any hypothetical width) in front of the luminaire (street side) and behind the luminaire (house side). The curves shall be marked as Street Side and House Side. Methods for computing utilization efficiency (luminous flux) may be found in **Annex D**.

6.5.5.5 Efficiency and Efficacy. Relative photometry: To calculate the total efficiency of the luminaire, the lumens in all areas shall be summed as described in **Annex A** and divided by the lumen rating of the lamp(s).

Absolute photometry: To calculate the total efficacy of the luminaire, the lumens in all areas shall be summed as described in **Annex A** and divided by the electrical power (in watts) measured during the photometric test.

Light output shall be expressed as a percentage of the lumen rating of the lamp (if appropriate) and as lumens subdivided into sections of a sphere as defined in the IES Luminaire Classification System (LCS)[14] (see **Figure 6-10**):

- Uplight
- Forward Light
- Back Light

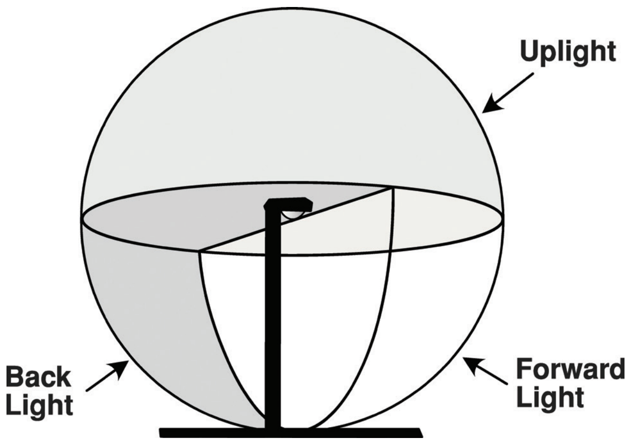


Figure 6-10. The three primary solid angles of the Luminaire Classification System (LCS).[14] (© Illuminating Engineering Society)

7.0 Test Report

7.1 General

The test report shall describe all significant data concerning the luminaire tested, and the data should be listed so that the maximum amount of information can be interpreted.[15] **Section 7.2** identifies data items to be included in the test report. The sample report in **Annex E** illustrates typical data and information content for luminous intensity, lumen, and lux (footcandle) values, rather than actual test values.

7.2 Test Description

Test reports shall include all data and descriptive information necessary to adequately describe the lamp or luminaire under test and the test conditions. The goal of the data is to provide users of the report with sufficient information to verify that the contents of the report are relevant to the planned use of both the report and the associated "IES file" data. While the exact information recorded is specific to the devices being tested, and the format is somewhat specific to the testing agency (except as specified in **Section 6**), the recorded data shall provide the following types of information:

1) Basic test information
   a) Test number
   b) Test date
   c) Test distance (light path distance from goniophotometer center to photodetector)
   d) Testing agency
2) Luminaire description
   a) Manufacturer's name; type and catalog number of luminaire, or indication of a prototype or test assembly
   b) Reflector description (e.g., material [surface finish, texture, specularity], reflectance [of painted surfaces], manufacturer's designation, dimensions)
   c) Enclosure or refractor description (e.g., material[s], manufacturer's designation, dimensions)
   d) Description of any auxiliary reflecting devices (e.g., louvers, shields)
   e) Ballast description (e.g., catalog number[s], number of lamps, power factor correction, voltage, frequency, lamp current if measurable, ballast location)
   f) Luminaire mounting and orientation (e.g., horizontal or tilted, angle of tilt)
3) Test lamp description
   a) Lamp catalog number, description, and number of lamps
   b) Electrical parameters (e.g., lamp watts, input watts, input voltage [where applicable])
   c) Rated lumen output at 25 °C (77 °F)
   d) Lamp orientation during test
4) Results—for roadway and area lighting
   (a) IES photometric data file[16]
   (b) Lumen output and efficiency of the luminaire, and total lumens on the house side and the street side

The recorded data should also provide the following optional types of information:

5) Luminaire description
   a) Graphic illustrations (e.g., housing and component shapes, dimensions, light center positions, goniophotometer center with respect to luminaire)
6) Test lamp description
7) Results—for roadway and area lighting

a) Luminous intensity distribution curves in the vertical plane and in the cone containing the maximum luminous intensity value
b) Luminous intensity tabulation
c) Horizontal lux (footcandle) lines of equal illuminance (isoilluminance in lux or footcandles) computed for an actual mounting height (if unknown, 9 m [or 30 ft] is recommended); **Annex F** shows the conversion of luminous intensity to horizontal illuminance values; this may be a tabulation of values in lieu of a diagram.
d) Street-side and House-side Coefficient of Utilization (tabular or graphical)
e) Zonal lumen sums and efficiencies of the lighting distribution in accordance with IES TM-15-11[14]
f) Classification of the lighting distribution in accordance IES area and roadway "transverse" (i.e., Types I, II, III, IV, V, and VS) and "longitudinal" classifications (i.e., Short, Medium, and Long)[2]

8) Miscellaneous other information
a) Temperature correction table or curve (see **Annex G**)
b) Other useful information (e.g., a photograph)
c) Method(s) of luminaire sample selection

Annex A – Zonal Constants for Conversion

The zonal constant is a factor by which the mean intensity emitted by a source of light in a given angular zone is multiplied to obtain the lumens in the zone.[17-21] These constants may also be used to calculate the lumens in any sector and thus the total lumens from a luminaire. Care should be taken to choose areas small enough to approach integration.

By multiplying the luminous intensity values found in each plane of measurement by the appropriate zonal constant, lumens for the corresponding segment of each zone are obtained. Addition of all lumen values corresponding to a given vertical angle of measurement will supply the lumens in the total zone represented.

For computer calculation, the zonal constants can be expressed by:

$$K = \frac{\pi(\cos\theta_1 - \cos\theta_1)(\phi_1 - \phi_2)}{180},$$

where:

θ_1 and θ_2 are vertical angles (in degrees) bounding the angular zone for a Type C coordinate system

ϕ_1 and ϕ_2 are lateral angles (in degrees) bounding the angular zone[3]

A simplified calculation may be made for applications where only the total lumens for a given vertical zone are of interest, such as when calculating bare lamp output. For each vertical angle, a weighted average of the test lamp's relative luminous intensity for all lateral angles shall be determined. The zonal constants when calculated for a total vertical zone can be calculated with the simplified equation:

$$K = 2\pi(\cos\theta_1 - \cos\theta_2)$$

Annex B – Photometry of Luminaires Utilizing Extra High Output Fluorescent Lamps

The light output of extra high output (EHO) lamps, generally having a current loading of 1.5 amperes or more, is sensitive to orientation, ambient temperature fluctuations, and drafts. This places considerable burden upon the photometric facilities if so-called "point source" photometry is desired. For luminaires utilizing the 2.4-meter (8-foot) fluorescent lamps, this then requires a test distance of 12 m (40 ft) all around the luminaire mounted in its normal operating position.

A method that will accomplish the desired results, and which lends itself to the space limitations encountered in the various laboratories, is the rotating-photocell short test distance method.[22] An alternative method using a rotating photocell is also available.[23]

Annex C – Procedure for Computing Isoilluminance Curves

The procedure for computing isoilluminance curves is as follows:

1) Calculate horizontal illuminance values for a series of vertical angles for each lateral angle through which a vertical luminous intensity distribution curve has been taken. Vertical angles are typically taken in 5-degree increments from nadir to 60 degrees, in 2½-degree increments from 60 to 80 degrees, and in 1-degree increments from 80 to 86 degrees. Horizontal illuminance produced by light emitted at angles above 86 degrees is generally insignificant. Horizontal illuminance can be expressed by the formula:

$$E = \frac{I_\theta\left(\cos^3\theta\right)}{MH^2},$$

where:

I_θ = luminous intensity at a vertical angle of θ degrees

MH = mounting height of the luminaire

This is commonly referred to as the cosine-cubed law.[5]

2) Plot horizontal illuminance values versus the tangent of the vertical angle. This may be computed on semi-log paper, or on rectangular coordinate paper if the coordinate (horizontal illuminance values) scale is changed each time that value goes below one-tenth full-scale value.

3) The isoilluminance diagram has as its scale a ratio of distance to mounting height, which is the tangent of the vertical angle. Radial lines may be drawn on the chart representing the intersection of the vertical planes through which luminous intensity values are recorded. Then, from the plotted curves of horizontal illuminance values versus tangents, desired horizontal illuminance values may be selected and plotted, and then values of equal illuminance may be joined by a smooth curve to form the isoilluminance diagram.

4) There are several computer graphical programs available that will facilitate the construction of isoilluminance diagrams following the general methodology described above. The input for these programs is a complete luminous intensity distribution array provided in standard IES format.[16]

Annex D – Computing Utilization Efficiency

To compute utilization efficiency, it is necessary to determine the portion of the luminaire's lumen output that actually strikes the pavement. There are several methods for completing this calculation. Two methods are presented in this Annex.

D.1 Method 1A

The Zonal Utilization Factors (ZUF) found in **Table D-1** and corresponding zonal lumens shall be multiplied and summed to compute Utilization Efficiency. Zonal lumens shall be calculated from the luminous intensity data at the center of each calculation zone as shown in **Annex A**.

$$\sum_{i=1}^{n}\sum_{j=1}^{m} ZUF(i,j) \times Lumens(i,j),$$

where:

i = current horizontal zone (such as 0° to 10°, centered around the luminous intensity value at 5° horizontal)

j = current vertical zone (such as 30° to 40°, centered around the luminous intensity value at 35° vertical)

n = the number of horizontal zones; for **Table D-1**, $n = 9$

m = the number of vertical zones; for **Table D-1**, $m = 9$

Table D-1 shows ZUF values for one quadrant. ZUF values for the other quadrants would be equivalent. Depending on the symmetry of the luminaire, zonal lumens may vary between quadrants.

Table D-1. Zonal Utilization Factors

Angular Zones		Horizontal Angles†								
	Vertical Angles†	5	15	25	35	45	55	65	75	85
Ratio 0.5*	**5**	1	1	1	1	1	1	1	1	1
	15	1	1	1	1	1	1	1	1	1
	25	0.6	0.7	0.9	1	1	1	1	1	1
	35	0	0	0	0.1	0.5	0.9	1	1	1
	45	0	0	0	0	0	0.1	0.8	1	1
	55	0	0	0	0	0	0	0.1	0.9	1
	65	0	0	0	0	0	0	0	0.4	1
	75	0	0	0	0	0	0	0	0	0.7
	85	0	0	0	0	0	0	0	0	0.3
Ratio 1.0*	**5**	1	1	1	1	1	1	1	1	1
	15	1	1	1	1	1	1	1	1	1
	25	1	1	1	1	1	1	1	1	1
	35	1	1	1	1	1	1	1	1	1
	45	0.5	0.6	0.7	1	1	1	1	1	1
	55	0	0	0	0.1	0.5	0.9	1	1	1
	65	0	0	0	0	0	0.1	0.6	1	1
	75	0	0	0	0	0	0	0	0.5	1
	85	0	0	0	0	0	0	0	0	0.5
Ratio 2.0*	**5**	1	1	1	1	1	1	1	1	1
	15	1	1	1	1	1	1	1	1	1
	25	1	1	1	1	1	1	1	1	1
	35	1	1	1	1	1	1	1	1	1
	45	1	1	1	1	1	1	1	1	1
	55	1	1	1	1	1	1	1	1	1
	65	0.3	0.4	0.6	0.8	1	1	1	1	1
	75	0	0	0	0	0.1	0.4	0.8		1
	85	0	0	0	0	0	0	0	0.3	0.7
Ratio 3.0*	**5**	1	1	1	1	1	1	1	1	1
	15	1	1	1	1	1	1	1	1	1
	25	1	1	1	1	1	1	1	1	1
	35	1	1	1	1	1	1	1	1	1
	45	1	1	1	1	1	1	1	1	1
	55	1	1	1	1	1	1	1	1	1
	65	1	1	1	1	1	1	1	1	1
	75	0.1	0.2	0.3	0.5	0.7	0.9	1	1	1
	85	0	0	0	0	0	0	0.2	0.5	0.8

Table D-1. Zonal Utilization Factors *(continued)*

Angular Zones		Horizontal Angles†								
	Vertical Angles†	5	15	25	35	45	55	65	75	85
Ratio 4.0*	5	1	1	1	1	1	1	1	1	1
	15	1	1	1	1	1	1	1	1	1
	25	1	1	1	1	1	1	1	1	1
	35	1	1	1	1	1	1	1	1	1
	45	1	1	1	1	1	1	1	1	1
	55	1	1	1	1	1	1	1	1	1
	65	1	1	1	1	1	1	1	1	1
	75	0.5	0.6	0.7	0.9	1	1	1	1	1
	85	0	0	0	0	0	0.2	0.4	0.6	0.9
Ratio 5.0*	5	1	1	1	1	1	1	1	1	1
	15	1	1	1	1	1	1	1	1	1
	25	1	1	1	1	1	1	1	1	1
	35	1	1	1	1	1	1	1	1	1
	45	1	1	1	1	1	1	1	1	1
	55	1	1	1	1	1	1	1	1	1
	65	1	1	1	1	1	1	1	1	1
	75	0.8	0.9	1	1	1	1	1	1	1
	85	0	0	0	0.1	0.2	0.3	0.5	0.7	0.9

Table notes:
* Lateral distance across the street as a ratio of the mounting height (MH).
† Angles shown are at the center of 10 degree zones.

D.2 Method 1B

Method 1B is the same as Method 1A, except that the zonal utilization factors are calculated, allowing the use of angular zones or mounting height ratios other than those shown in **Table D-1**.

An algorithm for computing roadway Zonal Utilization Factors (ZUFs) is provided below. For this method, the following terms will be used:

Ratio: The lateral distance (across the street) as a ratio of the mounting height. The ratio may be any non-negative number. The typical Utilization Efficiency chart uses ratios in the range of 0 to 5.

Angular Zone: The size of the zone for which the ZUF and lumens are being calculated. In **Table D-1**, the angular zones are 10° x 10°.

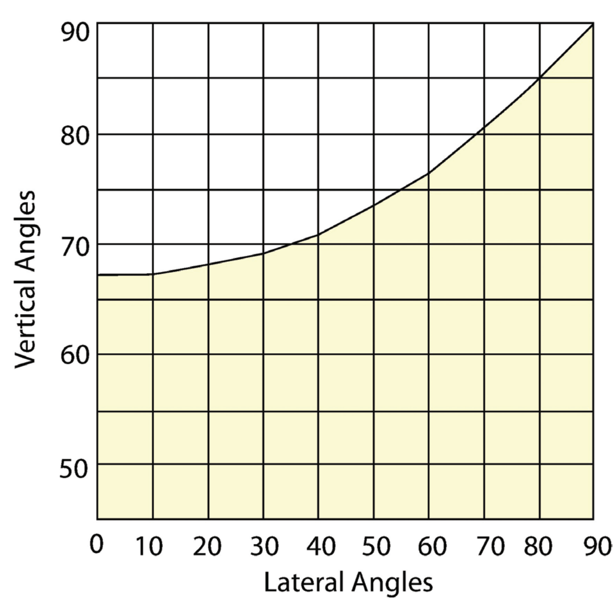


Figure D-1. Angular zones 10° x 10° for one quadrant. The shaded area illustrates the roadway passing through the angular zones. (© Illuminating Engineering Society)

For each ratio, it is necessary to calculate what portion of each angular zone falls on the roadway. In **Figure D-1**, for a ratio of 1.0, the shaded area would be on the roadway. Coverage is only shown for one quadrant, but would be the same for all four quadrants (two street-side and two house-side).

The Roadway ZUF is calculated from:

$Roadway\ ZUF = A_{polygon}/A_{square}$,

where:

$Roadway\ ZUF$ is the ZUF calculated for the roadway

$A_{polygon}$ is the area of the shaded polygon

A_{square} is the area of the square

For each angular zone, the ZUF is between 0 (no roadway in the zone) and 1 (zone completely on the roadway).

Each angular zone is bounded by four angles: V_{HI}, V_{LO}, H_{HI}, H_{LO}, as shown in **Figure D-2**.

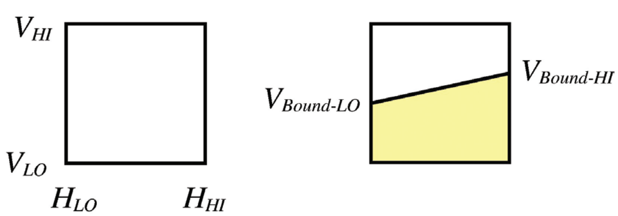


Figure D2-. Boundaries of one angular zone.
(© Illuminating Engineering Society)

For each of the two bounding horizontal angles (H_{HI}, H_{LO}), the vertical angle at which the edge of the roadway is intersected ($V_{\text{bound-HI}}$, $V_{\text{bound-LO}}$) is calculated:

$V_{\mathrm{bound}} = \tan^{-1}(Ratio/cosH)$

$If\ V_{\text{bound-HI}} \geq V_{\mathrm{HI}}$ and $V_{\text{bound-LO}} \geq V_{\mathrm{HI}}$, then $ZUF = 1$

$If\ V_{\text{bound-HI}} \leq V_{\mathrm{LO}}$ and $V_{\text{bound-LO}} \leq V_{\mathrm{LO}}$, then $ZUF = 0$

For all other conditions, determine the portion of the angular zone that falls on the roadway.

Figure D-3 shows the four scenarios that can occur. The shaded area falls on the roadway.

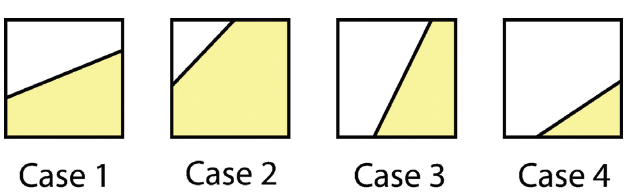


Figure D-3. Four possible cases for zones partially falling on the roadway. (© Illuminating Engineering Society)

Case 1:

$$A_{\text{polygon}} = \left(\frac{V_{\text{Bound}-HI} + V_{\text{Bound}-LO}}{2} - V_{LO}\right) \times (H_{\mathrm{HI}} - H_{\mathrm{LO}})$$

For Cases 2 through 4, calculation of H_{bound} values may be required in order to determine the coordinate of each polygon vertex:

$$H_{\text{Bound}} = \tan^{-1}\left[\frac{\sqrt{\tan V^2 - \tan(Ratio)^2}}{Ratio}\right]$$

Case 2:

$$A_{\text{polygon}} = A_{\text{square}} - \left(\frac{H_{\text{Bound–HI}} - H_{\mathrm{LO}}}{2}\right) \times (V_{\mathrm{HI}} - V_{\text{Bound–LO}})$$

Case 3:

$$A_{\text{polygon}} = \left(H_{\mathrm{HI}} - \frac{H_{\text{Bound–HI}} + H_{\text{Bound–LO}}}{2}\right)(V_{\mathrm{HI}} - V_{\mathrm{LO}})$$

Case 4:

$$A_{\text{polygon}} = \left(\frac{H_{\mathrm{HI}} - H_{\text{Bound–LO}}}{2}\right)(V_{\text{Bound–HI}} - V_{\mathrm{LO}})$$

D.3 Method 2

A second method makes use of the isoilluminance diagram.

Step 1.

On the isoilluminance diagram, squares are laid off whose sides are 0.5 MH by 0.5 MH or smaller. At the center of these squares, the horizontal illuminance value is read. Lumens in any square may then be calculated for the fundamental equation:

$Lumens$=*(Average luminous intensity)(Area)* ,

where:

Average luminous intensity is the horizontal value at the center of the Area

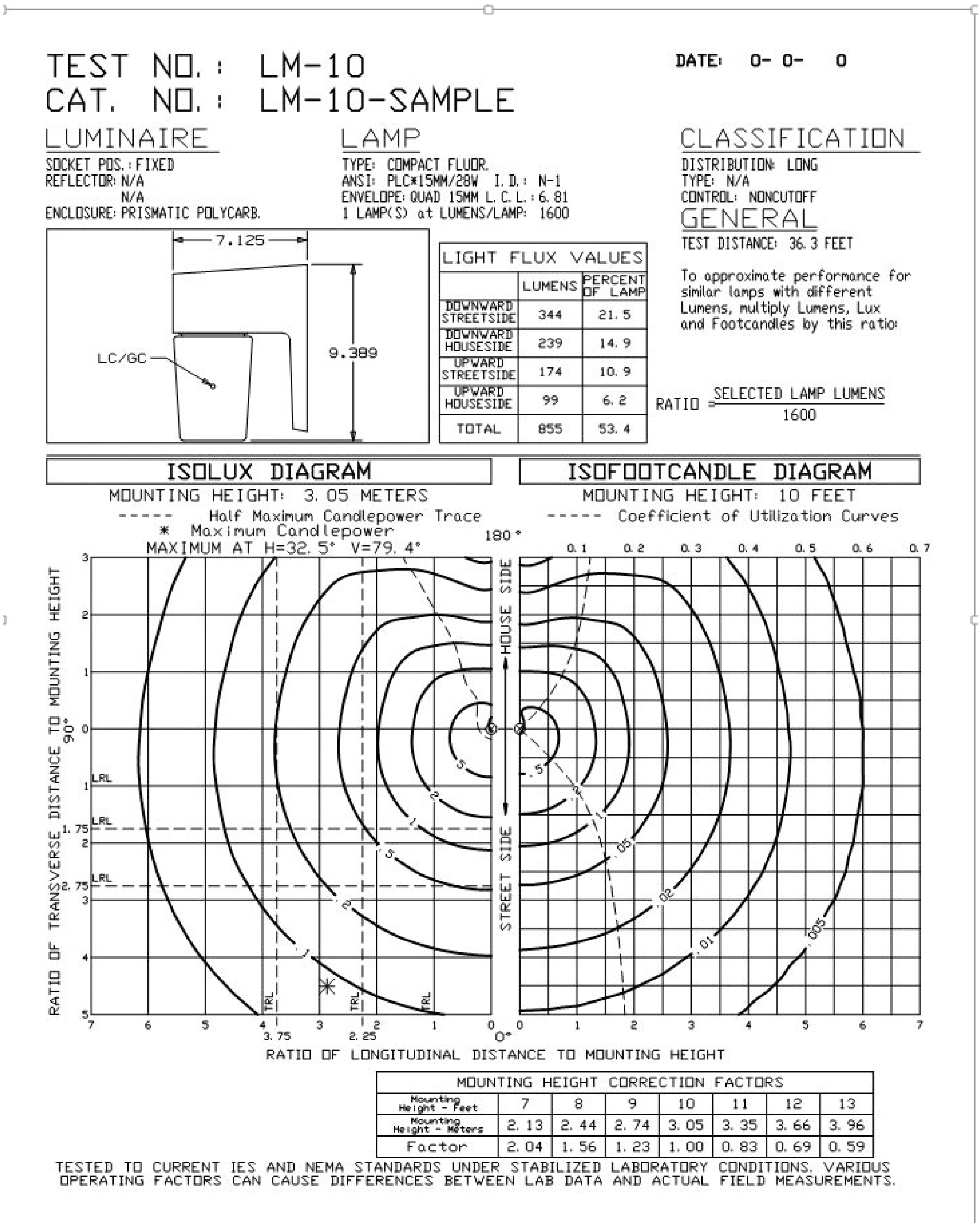
Area is the square of one-half the mounting height

Step 2.

Lumens in the square may then be summed in successive strips parallel to the curb line, each divided by the lumen rating used for the test lamp, and a plot of utilization versus transverse distance divided by mounting height may be made. It should be noted that if isoilluminance lines are shown for only one side of the axis of symmetry, the number of utilized lumen strips will double.

Annex E – Example Photometric Report

This Annex shows an example of a photometric report for a CFL area luminaire, first in graphical format, then in text format.


TEST NO.: LM-10
CAT. NO.: LM-10-SAMPLE

DATE: 0- 0- 0

LUMINAIRE
SOCKET POS.: FIXED
REFLECTOR: N/A
N/A
ENCLOSURE: PRISMATIC POLYCARB.

LAMP
TYPE: COMPACT FLUOR.
ANSI: PLC*15MM/28W I.D.: N-1
ENVELOPE: QUAD 15MM L.C.L.: 6.81
1 LAMP(S) at LUMENS/LAMP: 1600

CLASSIFICATION
DISTRIBUTION: LONG
TYPE: N/A
CONTROL: NONCUTOFF

GENERAL
TEST DISTANCE: 36.3 FEET

To approximate performance for similar lamps with different Lumens, multiply Lumens, Lux and Footcandles by this ratio:

RATIO = SELECTED LAMP LUMENS / 1600

LIGHT FLUX VALUES

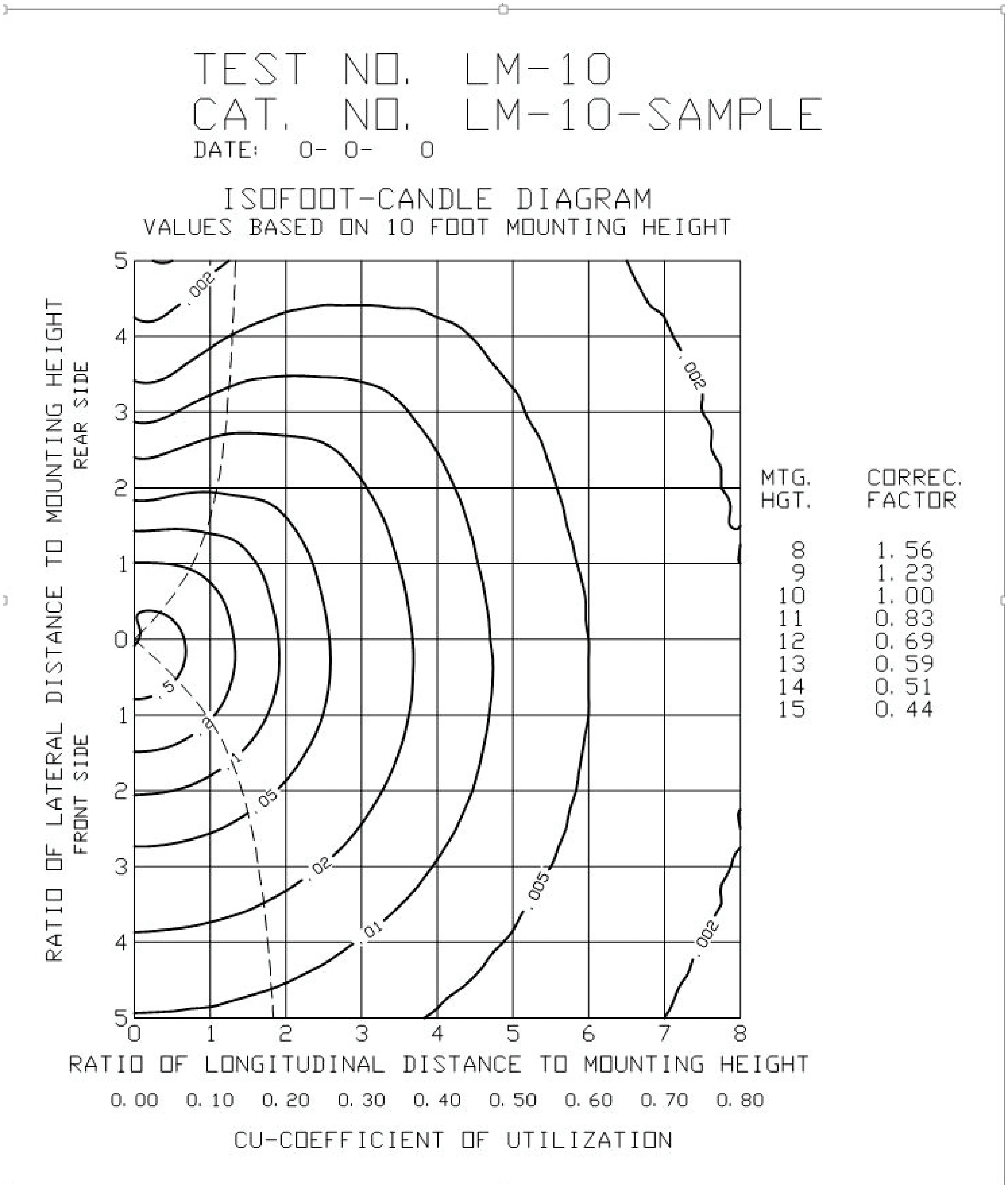
	LUMENS	PERCENT OF LAMP
DOWNWARD STREETSIDE	344	21.5
DOWNWARD HOUSESIDE	239	14.9
UPWARD STREETSIDE	174	10.9
UPWARD HOUSESIDE	99	6.2
TOTAL	855	53.4

ISOLUX DIAGRAM
MOUNTING HEIGHT: 3.05 METERS
----- Half Maximum Candlepower Trace
* Maximum Candlepower
MAXIMUM AT H=32.5° V=79.4°

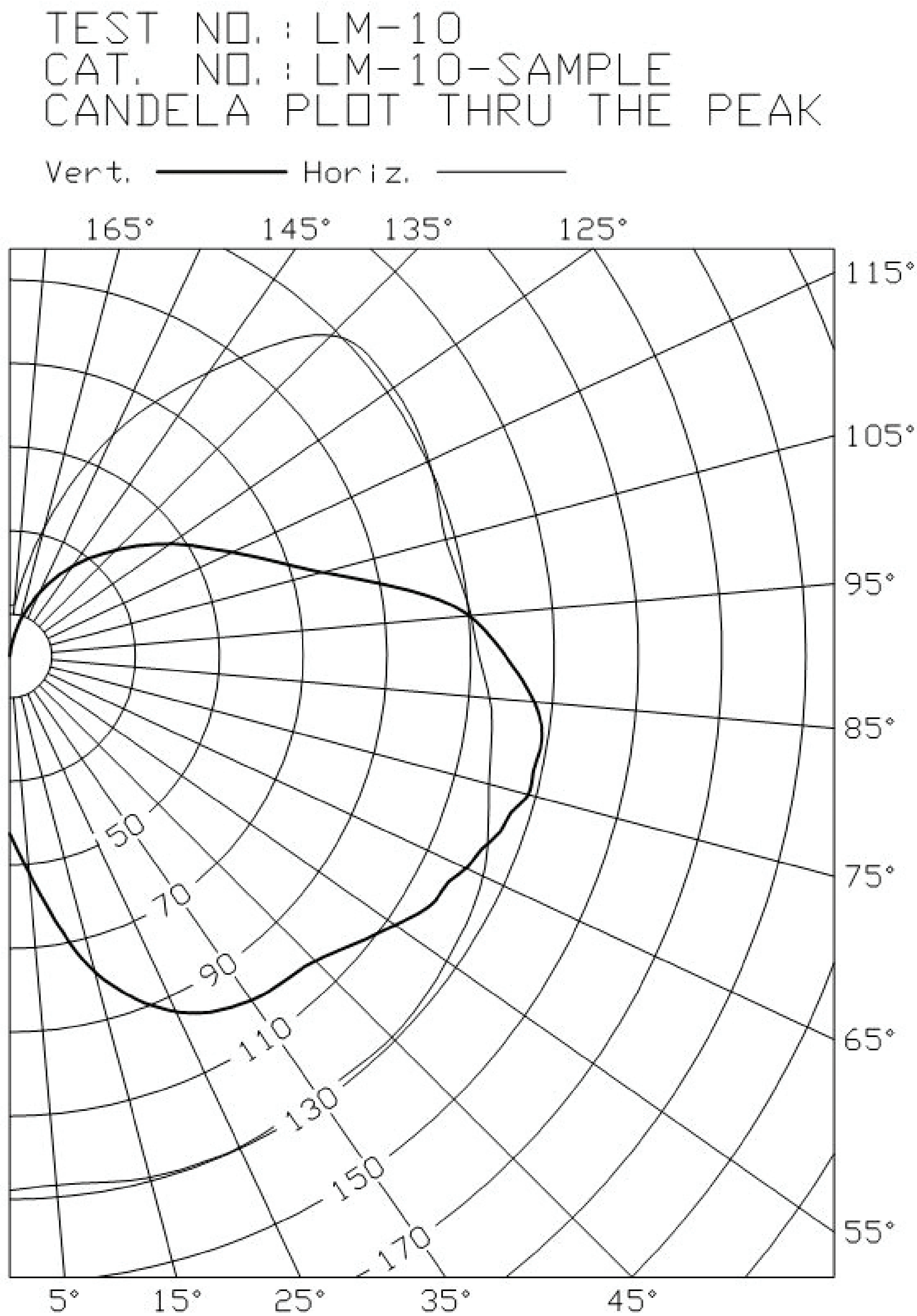
ISOFOOTCANDLE DIAGRAM
MOUNTING HEIGHT: 10 FEET
----- Coefficient of Utilization Curves

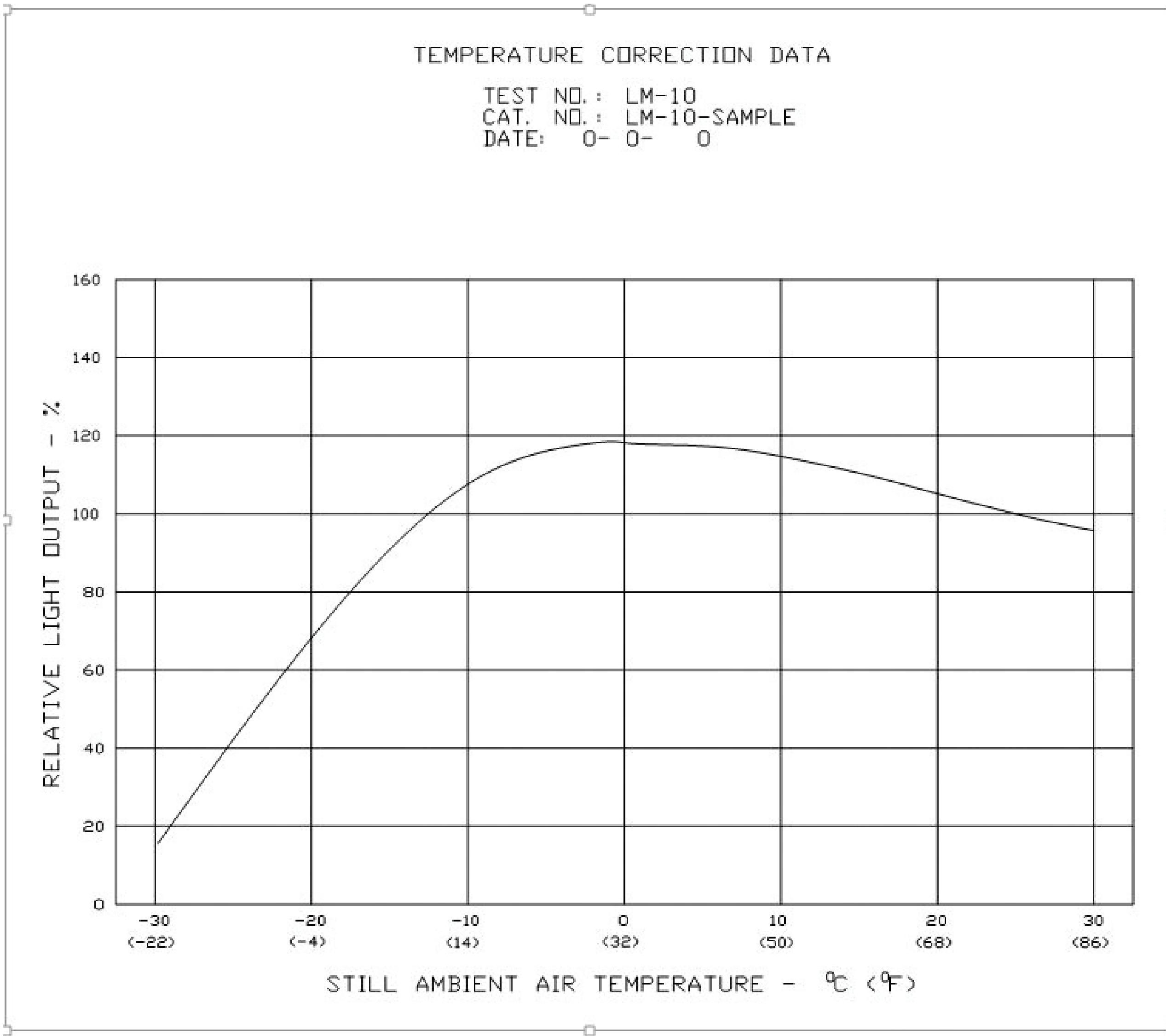
MOUNTING HEIGHT CORRECTION FACTORS							
Mounting Height - Feet	7	8	9	10	11	12	13
Mounting Height - Meters	2.13	2.44	2.74	3.05	3.35	3.66	3.96
Factor	2.04	1.56	1.23	1.00	0.83	0.69	0.59

TESTED TO CURRENT IES AND NEMA STANDARDS UNDER STABILIZED LABORATORY CONDITIONS. VARIOUS OPERATING FACTORS CAN CAUSE DIFFERENCES BETWEEN LAB DATA AND ACTUAL FIELD MEASUREMENTS.



TEST NO. LM-10
CAT. NO. LM-10-SAMPLE
DATE: 0- 0- 0
ISOFOOT-CANDLE DIAGRAM
VALUES BASED ON 10 FOOT MOUNTING HEIGHT
RATIO OF LATERAL DISTANCE TO MOUNTING HEIGHT
REAR SIDE
FRONT SIDE
5 4 3 2 1 0 1 2 3 4 5
.002
.002
.5
.2
.1
.05
.02
.01
.005
.002
0 1 2 3 4 5 6 7 8
RATIO OF LONGITUDINAL DISTANCE TO MOUNTING HEIGHT
0.00 0.10 0.20 0.30 0.40 0.50 0.60 0.70 0.80
CU-COEFFICIENT OF UTILIZATION
MTG. HGT. CORREC. FACTOR
8 1.56
9 1.23
10 1.00
11 0.83
12 0.69
13 0.59
14 0.51
15 0.44


TEST NO.: LM-10
CAT. NO.: LM-10-SAMPLE
CANDELA PLOT THRU THE PEAK
Vert.
Horiz.
165°
145°
135°
125°
115°
105°
95°
85°
75°
65°
55°
5°
15°
25°
35°
45°
50
70
90
110
130
150
170


TEMPERATURE CORRECTION DATA
TEST NO.: LM-10
CAT. NO.: LM-10-SAMPLE
DATE: 0- 0- 0
RELATIVE LIGHT OUTPUT - %
160
140
120
100
80
60
40
20
0
-30 (-22)
-20 (-4)
-10 (14)
0 (32)
10 (50)
20 (68)
30 (86)
STILL AMBIENT AIR TEMPERATURE - °C (°F)

REPORT NO. LM-10 DATE 0- 0- 0

CAT. NO. LM-10-SAMPLE

TESTED WITH LAMP TYPE PLC*15MM/28W

CANDELA

NADIR C.P. = 43., MAX. C.P. = 132.

LATERAL ANGLE

		0.0	5.0	15.0	25.0	35.0	45.0	55.0	65.0	75.0	85.0
	0.0	43.	43.	43.	43.	43.	43.	43.	43.	43.	43.
	5.0	53.	53.	53.	53.	53.	52.	51.	50.	49.	48.
	15.0	78.	78.	79.	79.	80.	80.	79.	77.	75.	72.
	25.0	94.	94.	94.	95.	95.	95.	93.	92.	88.	85.
	35.0	102.	101.	101.	102.	100.	99.	97.	95.	90.	87.
	45.0	106.	106.	106.	104.	103.	101.	98.	95.	91.	87.
	55.0	115.	115.	114.	114.	112.	110.	107.	103.	99.	94.
	60.0	118.	118.	119.	117.	117.	116.	113.	109.	104.	99.
V	62.5	118.	119.	119.	120.	119.	119.	115.	111.	106.	102.
E	65.0	121.	120.	121.	121.	120.	119.	117.	114.	109.	104.
R	67.5	122.	122.	122.	123.	122.	122.	118.	116.	111.	107.
T	70.0	124.	124.	125.	125.	124.	124.	122.	119.	114.	109.
I	72.5	125.	125.	127.	127.	126.	126.	124.	119.	116.	111.
C	75.0	128.	127.	127.	129.	128.	128.	126.	122.	118.	113.
A	77.5	128.	128.	129.	129.	129.	130.	127.	125.	118.	114.
L	80.0	128.	127.	129.	130.	129.	130.	129.	125.	119.	115.
	82.5	127.	127.	129.	129.	129.	129.	129.	124.	120.	115.
A	85.0	126.	125.	127.	127.	126.	129.	127.	124.	119.	114.
N	90.0	120.	118.	119.	120.	119.	122.	120.	117.	112.	107.
G	95.0	109.	109.	110.	111.	111.	112.	110.	107.	103.	98.
L	105.0	75.	77.	77.	78.	77.	78.	76.	75.	72.	69.
E	115.0	56.	57.	58.	59.	59.	59.	58.	56.	54.	51.
	125.0	45.	46.	46.	47.	47.	47.	47.	45.	43.	41.
	135.0	34.	35.	35.	36.	36.	36.	36.	35.	33.	31.
	145.0	26.	26.	26.	27.	27.	27.	26.	26.	25.	24.
	155.0	17.	17.	17.	17.	17.	17.	16.	16.	15.	14.
	165.0	8.	8.	8.	8.	8.	8.	8.	8.	7.	7.
	175.0	1.	1.	1.	1.	1.	1.	1.	1.	1.	1.
	180.0	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.

REPORT NO. LM-10 DATE 0- 0- 0

CAT. NO. LM-10-SAMPLE

TESTED WITH LAMP TYPE PLC*15MM/28W

CANDELA

NADIR C.P. = 43., MAX. C.P. = 132.

LATERAL ANGLE

		90.0	95.0	105.0	115.0	125.0	135.0	145.0	155.0	165.0	175.0
	0.0	43.	43.	43.	43.	43.	43.	43.	43.	43.	43.
	5.0	47.	46.	45.	44.	43.	42.	42.	41.	40.	40.
	15.0	71.	70.	68.	65.	63.	61.	59.	57.	55.	54.
	25.0	83.	82.	79.	75.	72.	70.	67.	64.	61.	60.
	35.0	85.	84.	80.	77.	74.	71.	68.	66.	63.	63.
	45.0	85.	83.	81.	78.	76.	73.	69.	65.	60.	58.
	55.0	92.	91.	90.	90.	88.	84.	75.	67.	59.	53.
	60.0	97.	96.	95.	96.	96.	91.	78.	67.	56.	48.
V	62.5	100.	98.	96.	99.	99.	94.	79.	67.	54.	44.
E	65.0	102.	100.	99.	101.	101.	97.	80.	67.	51.	40.
R	67.5	104.	103.	101.	103.	104.	100.	80.	65.	48.	35.
T	70.0	106.	105.	103.	105.	106.	103.	81.	64.	44.	30.
I	72.5	109.	108.	105.	107.	108.	104.	82.	62.	40.	25.
C	75.0	110.	109.	107.	109.	110.	107.	82.	60.	37.	20.
A	77.5	111.	109.	107.	110.	111.	108.	82.	58.	33.	16.
L	80.0	112.	111.	108.	110.	111.	108.	82.	56.	30.	12.
	82.5	112.	110.	107.	110.	112.	108.	82.	54.	27.	9.
A	85.0	110.	108.	106.	108.	110.	107.	80.	52.	25.	6.
N	90.0	104.	102.	100.	102.	104.	102.	75.	48.	22.	3.
G	95.0	95.	93.	92.	95.	97.	94.	69.	43.	18.	1.
L	105.0	66.	66.	66.	69.	72.	69.	46.	28.	12.	1.
E	115.0	49.	48.	48.	49.	48.	43.	29.	18.	8.	1.
	125.0	39.	38.	37.	35.	31.	26.	18.	11.	5.	1.
	135.0	29.	29.	26.	23.	19.	15.	11.	7.	3.	0.
	145.0	22.	22.	19.	16.	13.	10.	8.	5.	2.	0.
	155.0	14.	13.	12.	10.	8.	6.	4.	2.	1.	0.
	165.0	6.	6.	5.	4.	3.	2.	1.	1.	0.	0.
	175.0	1.	0.	0.	0.	0.	0.	0.	0.	0.	0.
	180.0	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.

```
REPORT NO. LM-10                                    DATE  0- 0- 0
CAT. NO. LM-10-SAMPLE
TESTED WITH LAMP TYPE  PLC*15MM/28W
CANDELA
NADIR C.P. =  43., MAX. C.P. = 132.
                          LATERAL ANGLE
             180.0
       0.0     43.
       5.0     40.
      15.0     54.
      25.0     60.
      35.0     63.
      45.0     58.
      55.0     53.
      60.0     47.
   V  62.5     44.
   E  65.0     39.
   R  67.5     35.
   T  70.0     30.
   I  72.5     25.
   C  75.0     20.
   A  77.5     15.
   L  80.0     11.
      82.5      8.
   A  85.0      6.
   N  90.0      3.
   G  95.0      1.
   L 105.0      1.
   E 115.0      1.
     125.0      1.
     135.0      0.
     145.0      0.
     155.0      0.
     165.0      0.
     175.0      0.
     180.0      0.
```

REPORT NO. LM-10 DATE 0- 0-

CAT. NO. LM-10-SAMPLE

TESTED WITH LAMP TYPE PLC*15MM/28W

CONSTANTS

LAMP CONSTANT	157.921
NUMBER OF LAMPS	1
VENDOR LUMENS/LAMP	1600.
PER UNIT M.H.	10.0
TOTAL LUMENS	855.
TOTAL EFFICIENCY	53.4

LUMENS BY VERTICAL QUADRANTS

	STREET SIDE		HOUSE SIDE	
	LUMENS	PCT	LUMENS	PCT
DOWN	344.	21.5	239.	14.9
UP	174.	10.9	98.	6.2

UTILIZATION TABLE

	STREET SIDE		HOUSE SIDE	
MOUNTING	UTILIZED		UTILIZED	
HEIGHT RATIO	LUMENS	EFF.	LUMENS	EFF.
0.000	0.0	0.0	0.0	0.0
0.500	84.8	5.3	72.2	4.5
1.000	152.3	9.5	122.9	7.7
1.500	197.2	12.3	154.6	9.7
2.000	227.0	14.2	174.5	10.9
2.500	247.5	15.5	187.3	11.7
3.000	262.2	16.4	196.1	12.3
4.000	281.7	17.6	207.1	12.9
5.000	293.9	18.4	213.6	13.4
(INTERPOLATED VALUES)				
0.250	44.6	2.8	38.8	2.4
0.750	120.7	7.5	100.2	6.3
1.250	177.5	11.1	141.1	8.8
1.750	214.0	13.4	166.0	10.4
2.250	238.4	14.9	181.8	11.4
2.750	255.6	16.0	192.2	12.0

3.250	268.4	16.8	199.6	12.5
3.500	273.7	17.1	202.7	12.7

BUG Rating: B0 U3 G1

Luminaire Classification System (LCS)

	Lumens	%Lamp
FRONT LOW (0-30)	22.6	2.6
FRONT MED (30-60)	129.1	15.1
FRONT HIGH (60-80)	123.2	14.4
FRONT VERY HIGH (80-90)	67.1	7.9
BACK LOW (0-30)	17.8	2.1
BACK MED (30-60)	94.1	11.0
BACK HIGH (60-80)	83.9	9.8
BAC VERY HIGH (80-90)	42.4	5.0
UPLIGHT LOW (90-100)	50.0	5.9
UPLIGH HIGH (100-180)	223.5	26.2
TOTAL	853.5	

NOTES:

-TESTED PER IESNA LM-10 (OUTDOOR FLUORESCENT) GUIDELINES.

-LCS DATA PER IESNA TM-15

Annex F – Table of Constants for Horizontal Illuminance

This Annex provides pre-calculated constants that can be multiplied by luminaire luminous intensity (I) to obtain horizontal illuminance (E).

$$E = I \times C_E ,$$

where:

$C_E = \cos^3(Vertical\ Angle)/(Mounting\ Height)^2$ and can be obtained from **Table F-1**

E = Horizontal illuminance at the target location

I = Luminous Intensity in the direction of the target

Vertical Angle = the vertical angular deflection to the target

Table F-1. Table of Constants for Horizontal Illuminance (C_E)

	Mounting Height (Meters or Feet)			
Vertical Angle	6	7.5	9	Tangent
0	0.02778	0.01778	0.01235	0.000
5	0.02746	0.01758	0.01221	0.087
10	0.02653	0.01698	0.01179	0.176
15	0.02503	0.01602	0.01113	0.268
20	0.02305	0.01475	0.01024	0.364
25	0.02068	0.01323	0.009191	0.466
30	0.01804	0.01155	0.008019	0.577
35	0.01527	0.009772	0.006786	0.700
40	0.01249	0.007992	0.005550	0.839
45	0.009821	0.006285	0.004365	1.000
50	0.007377	0.004721	0.003279	1.192
55	0.005242	0.003355	0.002330	1.428
60	0.003472	0.002222	0.001543	1.732
65	0.002097	0.001342	0.0009319	2.145
70	0.001111	0.0007113	0.0004939	2.747
75	0.0004816	0.0003082	0.0002140	3.732
78	0.0002497	0.0001598	0.0001110	4.705
80	0.0001454	0.00009309	0.00006464	5.671
81	0.0001063	0.00006806	0.00004726	6.314
82.5	0.00006177	0.00003953	0.00002745	7.596
83	0.00005028	0.00003218	0.00002235	8.144
84	0.00003172	0.00002030	0.00001410	9.514
85	0.00001839	0.00001177	0.000008173	11.430
86	0.000009429	0.000006034	0.000004191	14.301

Annex G – Temperature Correction Factors

To obtain the temperature correction curves described here, the temperature-controlled enclosure shall be large enough to house the luminaire. There shall be minimum movement of air about the luminaire during the stabilization and reading of the light output at a given luminaire ambient temperature; however, this does not necessarily imply that there is anything undesirable about the normal convection currents because of the heat of the luminaire.

Instrumentation should be outside the temperature enclosure to avoid errors in readings due to possible thermal effects on the instrument's readings.

If these conditions can be satisfactorily established, the next step is to measure the relative light output of the luminaire in percent along with the bulb wall temperature of the lamp in the luminaire, over an ambient temperature range of approximately 40 °C to -30 °C. From the resulting data, curves of relative light output in percent vs. bulb wall temperature are plotted, as well as bulb wall temperature in the luminaire vs. ambient temperature:

Step 1:
Given: Average bulb wall temperature with the luminaire in normal operating position.

Find: Light output.

Solution: From **Figure G-1(a)**, read the light output for given bulb wall temperature. Call this value *a*.

Step 2:
Given: Ambient temperature at time of test.

Find: Light output.

Solution: From **Figure G-1(b)**, read bulb wall temperature for given ambient. From **Figure G-1(a)**, read corresponding light output for bulb wall temperature found. Call this value *b*.

Step 3:
Given: Reference ambient temperature: 25 °C (77 °F).

Find: Light output.

Solution: From **Figure G-1(b)**, read bulb wall temperature for 25 °C (77 °F) ambient. From **Figure G-1(a)**, read corresponding light output for bulb wall temperature found. Call this value *d*.

Step 4:
Compute correction factor, C:

$$C = (X+Y)/2 \,,$$

where:

$$X = 100\frac{d}{b}$$

$$Y = 100\frac{d}{a}$$

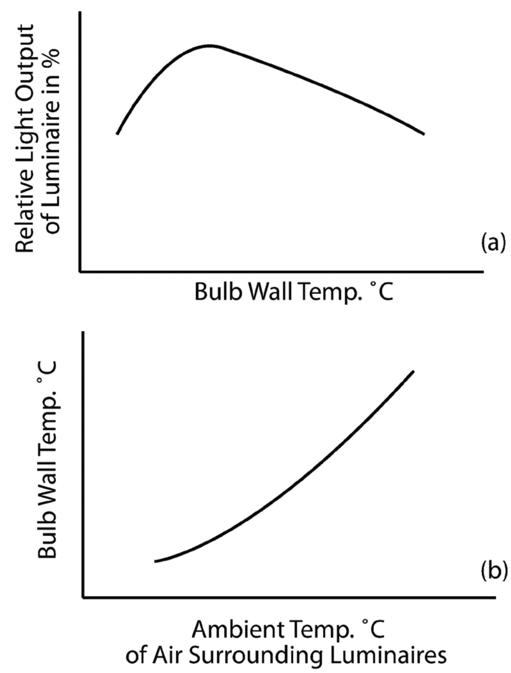


Figure G--1. Typical curves for temperature correction factors: (a) light output versus bulb wall temperature for lamps in luminaire; and (b) bulb wall temperature versus ambient temperature for lamps in the luminaire. Specific curves for specific tests shall have graduated scales.
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